

Differential pair rules in gastrointestinal aGVHD

A visual catalogue of method checks, atlas-linked results, and new candidates

525 directional pair rules; 267 FOVs; 262 with a clinical severity score. Arrows describe conditional association, not causation.

Automatic severity selection ranks Control–Severe attraction and avoidance gaps separately, requiring at least 10 eligible FOVs at both endpoints and a change of at least 20 percentage points. A second screen ranks biopsy-tested effects and large visible effects separately. Findings without adjusted biopsy evidence are labelled exploratory. Abundance appears only as a control.

1 Method proof

Epithelial–goblet and vascular rules recur across stages, as expected for familiar anatomy.

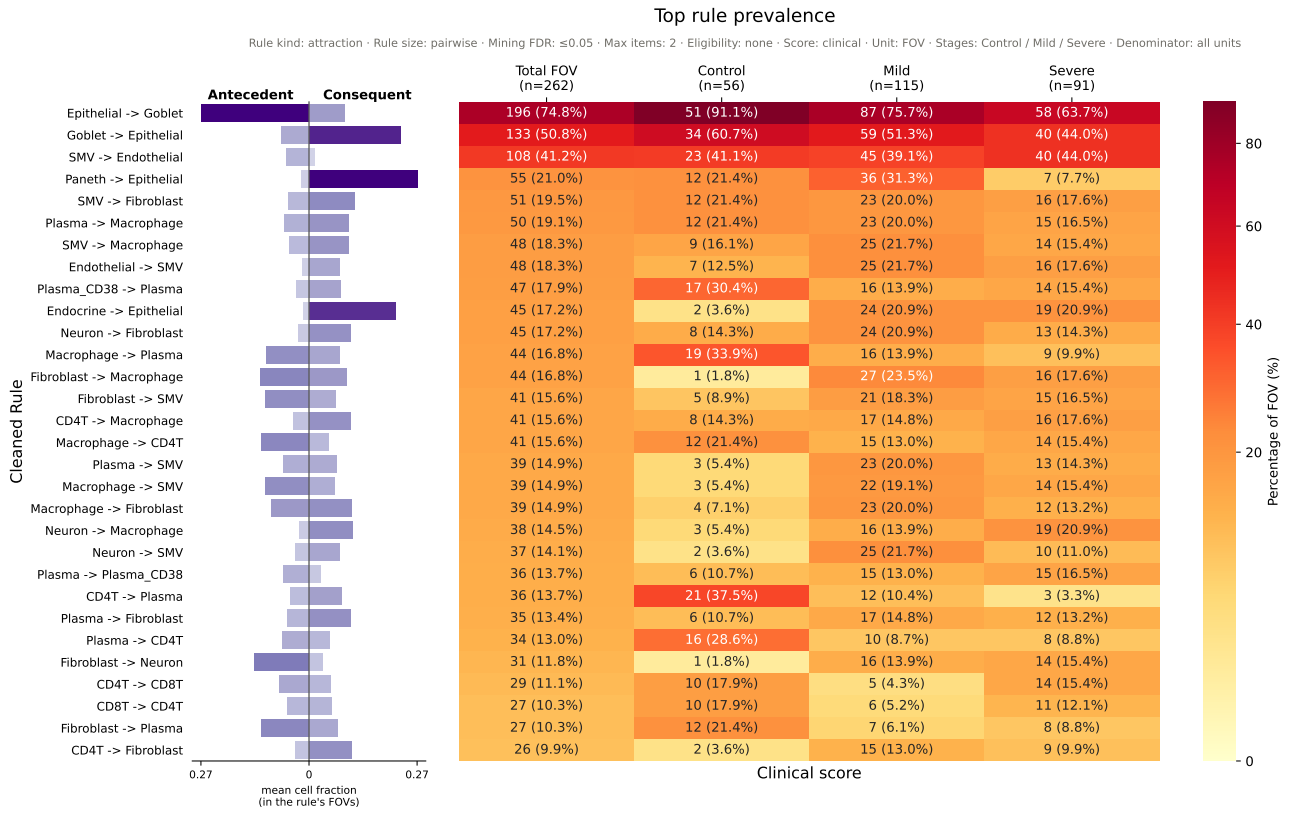


Figure 1: Rows are the most frequent mined attraction rules. Heatmap cells give rule-bearing FOVs over all FOVs in each stage; the bars at left give the two cell-type fractions in rule-bearing FOVs. This overview has no 20-cell eligibility filter and uses the 262 FOVs with a clinical score.

The same anatomical rules appear in many patients, beyond individual tissue fields.

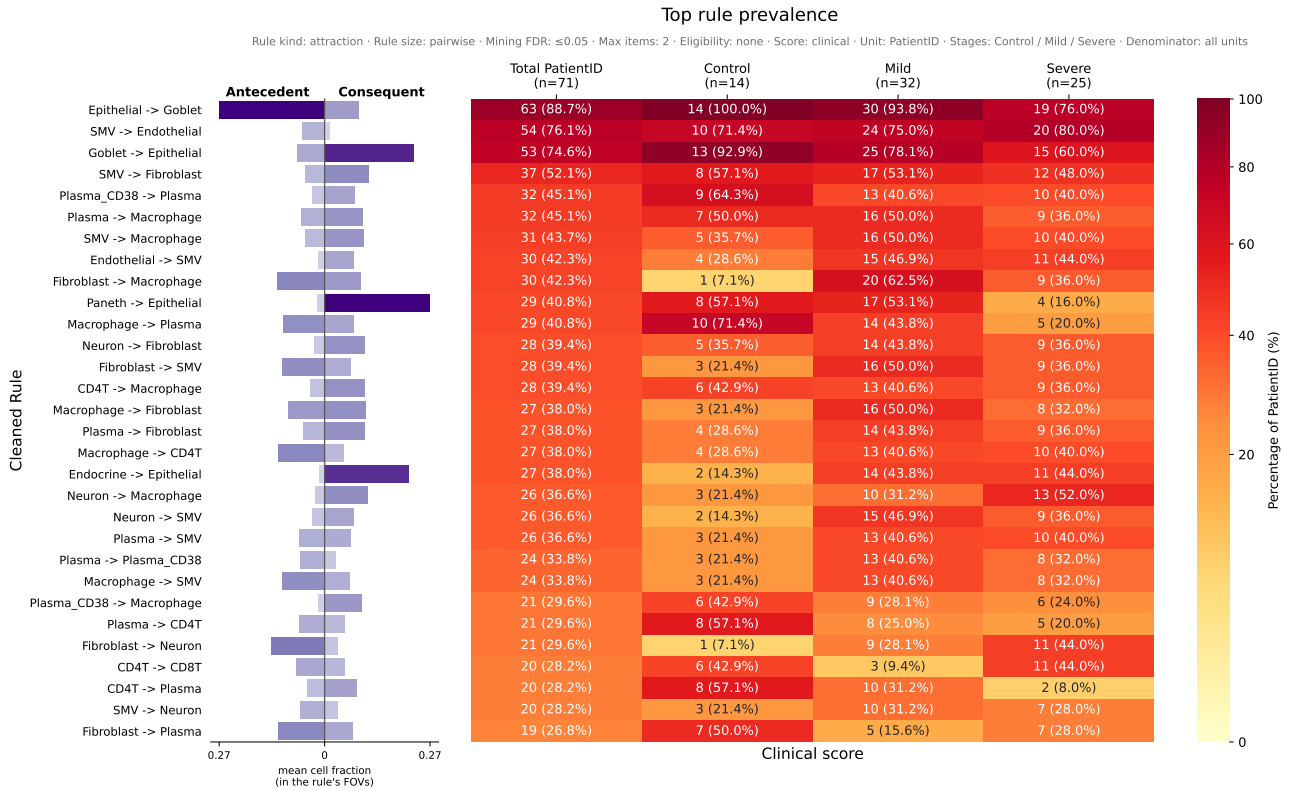


Figure 2: As above, with one count per patient if any of their FOVs contains the rule. Patient counts show recurrence; they are not tests of a stage difference.

Epithelial → Goblet stays attractive in eligible fields, consistent with their shared intestinal architecture [2].

Colon · Epithelial → Goblet

Plots: FOV · Tests: biopsy · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs

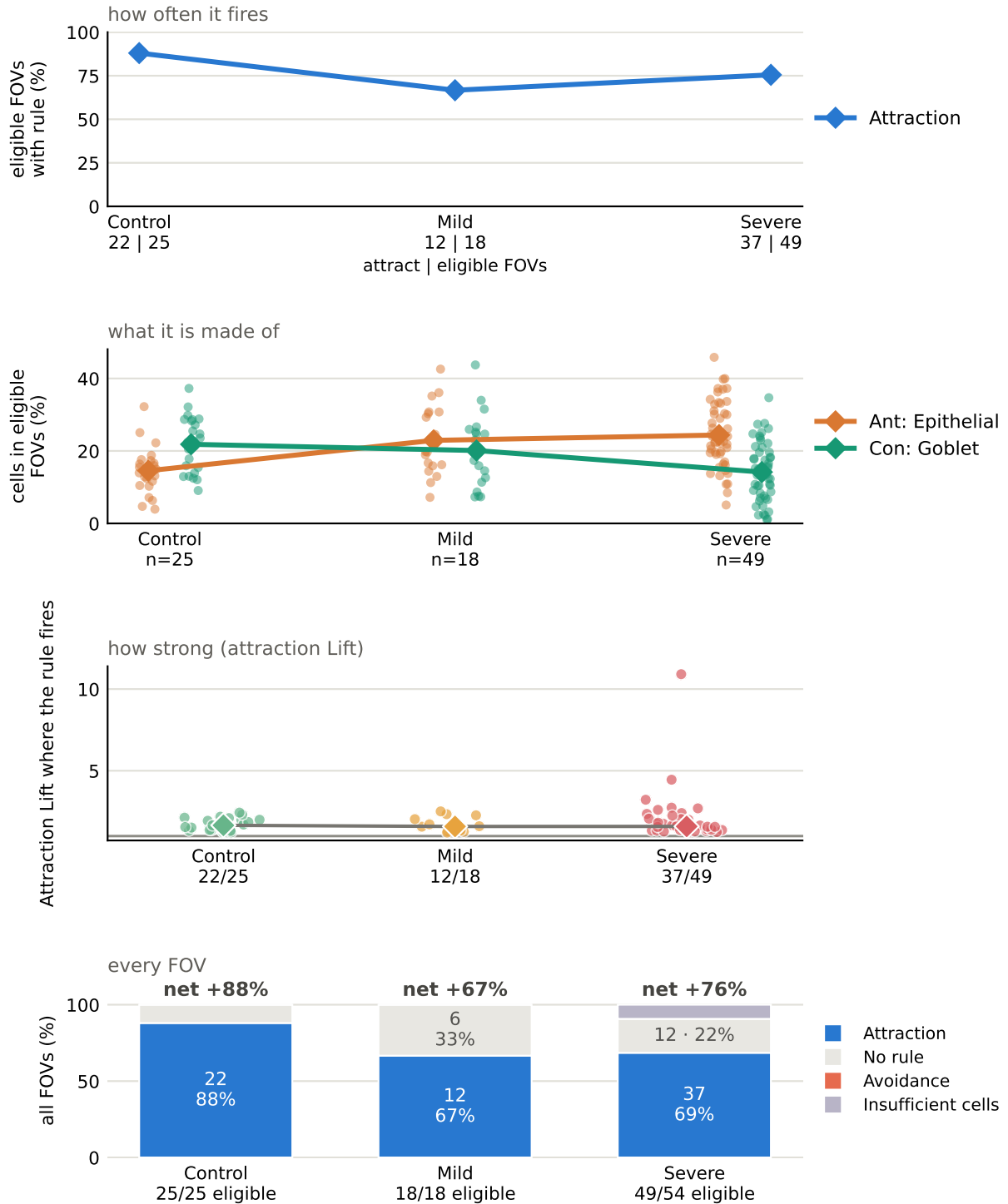


Figure 3: This is the first shared rule summary. The four rows use the same rule and clinical stages. **How often it fires:** attraction and avoidance are percentages of eligible FOVs; the numbers below each stage are raw attraction | avoidance | eligible counts, with an all-zero direction omitted. **What it is made of:** dots are the two cell-type shares inside each eligible FOV and diamonds are stage means; this is an abundance control, not a finding. **How strong:** dots are Lift in rule-bearing FOVs and diamonds are medians; Lift 1 is chance. **Every FOV:** bars include attraction, no rule, avoidance, and insufficient-cell fields, with net score above. These FOV percentages are descriptive; fields from one patient are clustered. An FOV is eligible only when it contains at least 20 cells of both rule types. Its state is attraction, avoidance, or no rule; net score is $100(N_{\text{attraction}} - N_{\text{avoidance}})/N_{\text{eligible}}$. The biopsy-level gap screen does not support a severity change in Lift here (FDR $q = 0.660$).

SMV → Endothelial stays attractive across stages, recovering a vascular relationship.

Colon · SMV → Endothelial

Plots: FOV · Tests: biopsy · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs

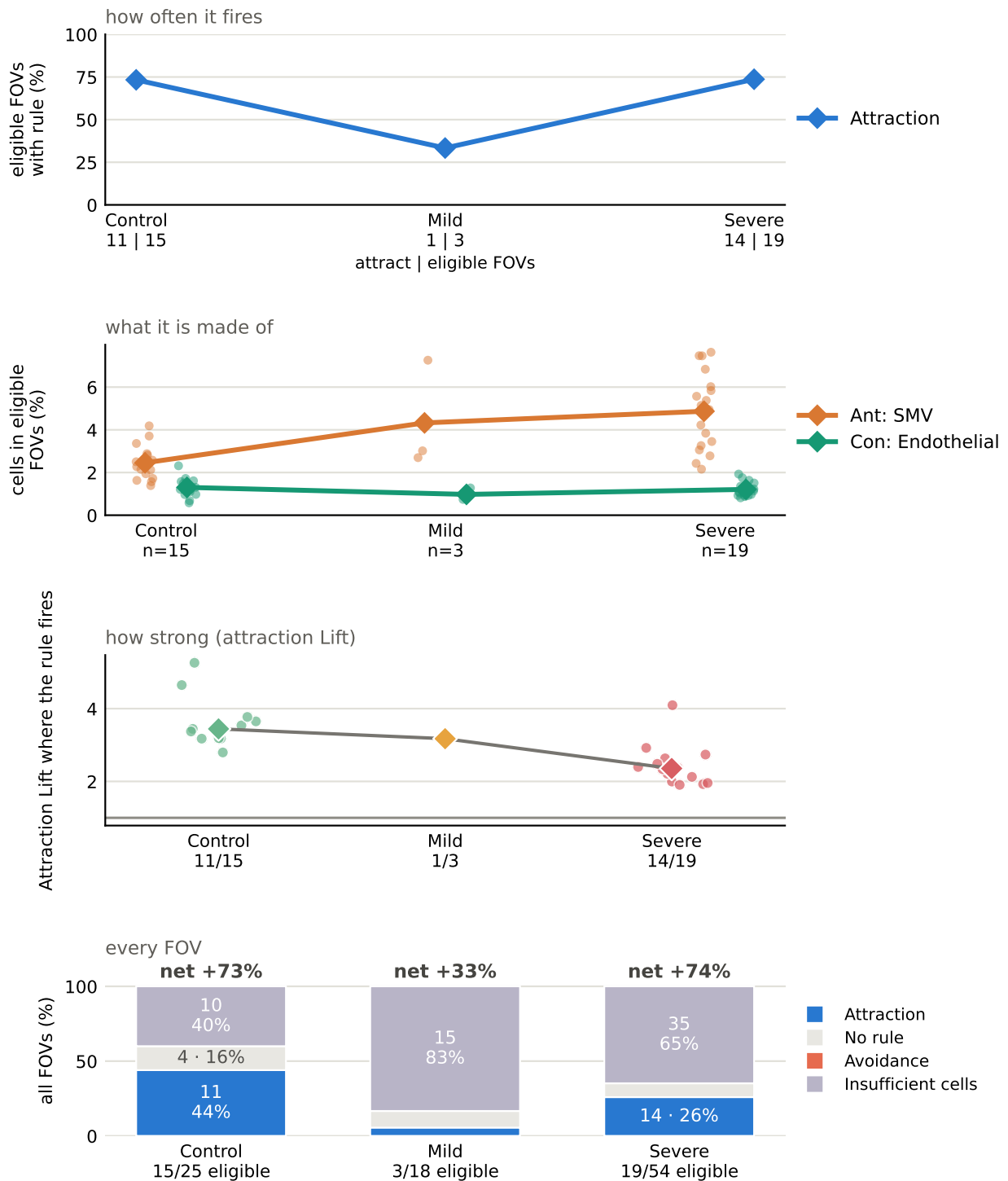


Figure 4: Read as in Figure 3. This anatomical rule is recurrent, while its biopsy-level Lift difference by stage is not supported (FDR $q = 0.370$).

Plasma_CD38 → Plasma stays strongly attractive, providing a check using related plasma-cell phenotypes.

Duodenum · Plasma_CD38 → Plasma

Plots: FOV · Tests: biopsy · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs

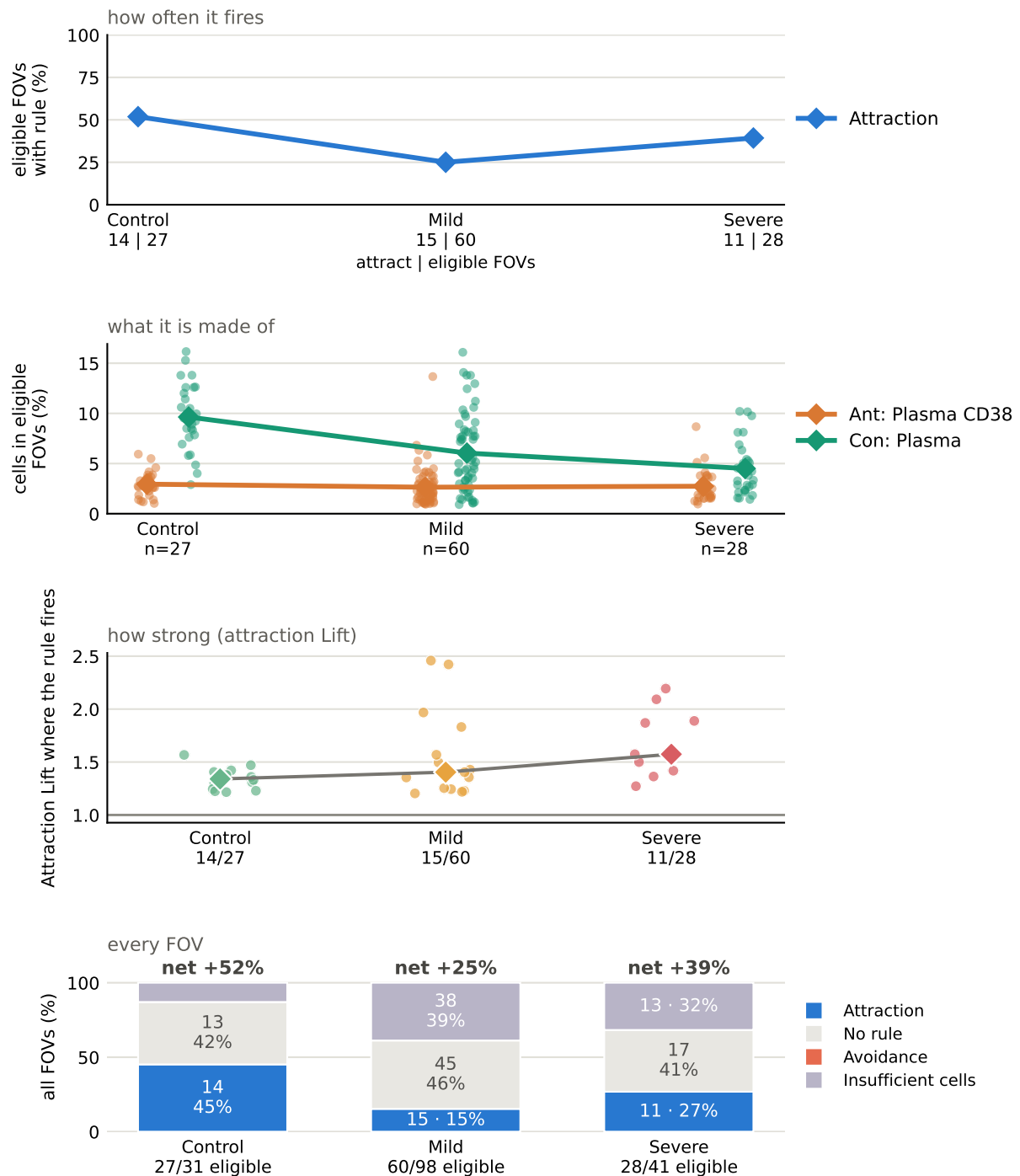


Figure 5: Read as in Figure 3. The biopsy-level Lift screen gives FDR $q = 0.224$ for stage change; this figure is a related-phenotype method check.

1.1 Known biological pairs tested in both directions

Six literature-selected pairs are checked retrospectively; the notebook shows both directions, whole-organ prevalence, stage-specific results and typical tissue examples.

SMV → Endothelial attracts in 26/37 eligible colon fields, recovering an expected vessel-wall relationship [8].

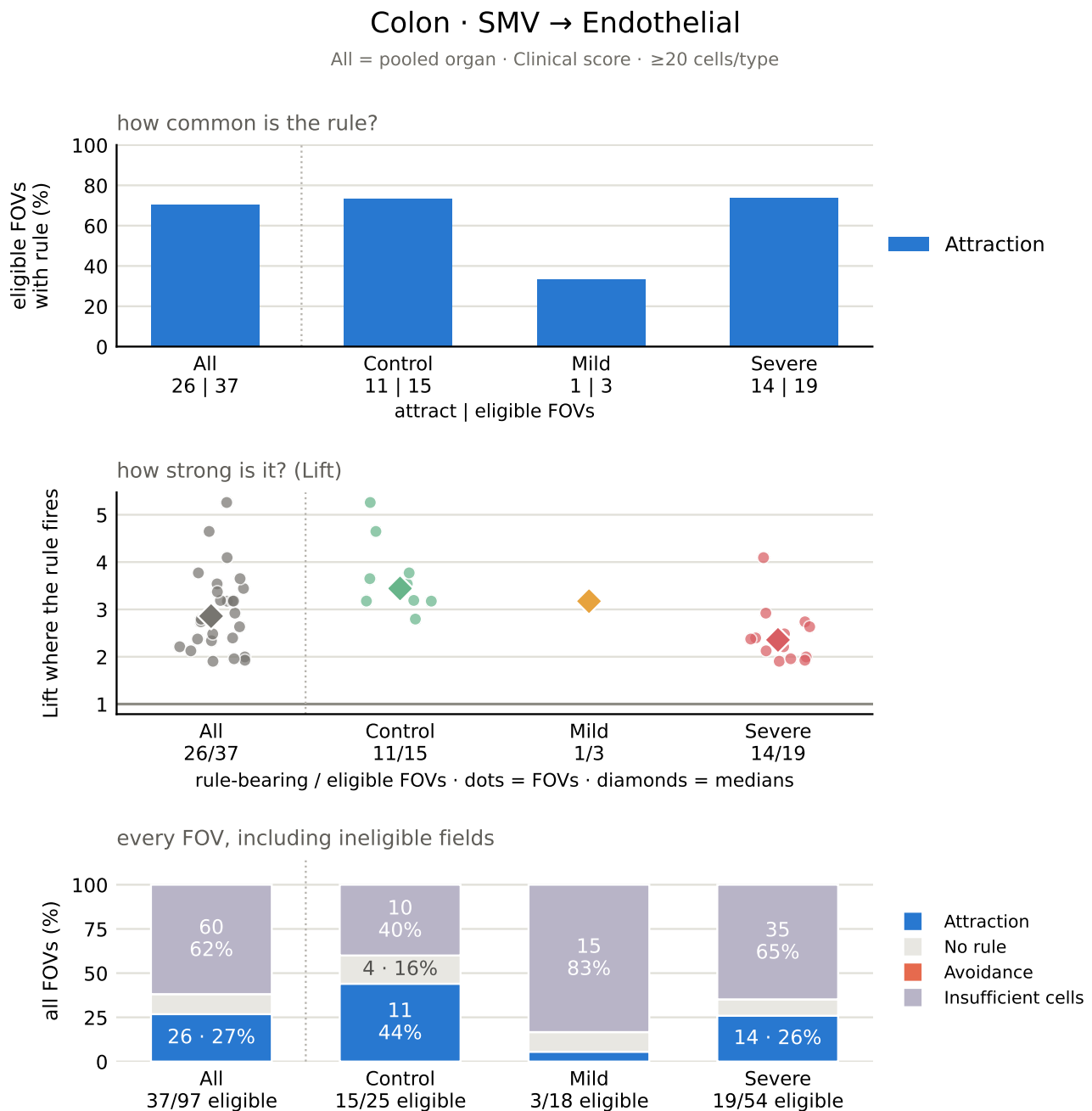


Figure 6: Both directions use the same FOVs with at least 20 cells of each type. All pools the organ's fields; the other columns split those same fields by clinical stage. Top: attraction and avoidance counts divided by eligible FOVs; counts below are attract | avoid | eligible, omitting an all-zero state. Middle: raw mined Lift in passing, eligible FOVs; dots are FOVs, diamonds medians, and Lift 1 is the independence baseline. Missing rules have no plotted Lift. Bottom: all FOVs, separated into attraction, avoidance, eligible without a rule, and insufficient cells. Patient q values below compare the two arrows by a two-sided sign test after averaging FOVs within biopsies and biopsies within patients; FDR uses the notebook's joint 119-test family. The same eligibility filter is applied to the reverse arrow. Here attraction occurs in 26/37 fields versus 12/37 in reverse; the patient sign test gives FDR $q = 0.0021$.

The highlighted vascular cells show where the recurrent colon rule sits within whole fields.

Colon · SMV → Endothelial

Attraction examples

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

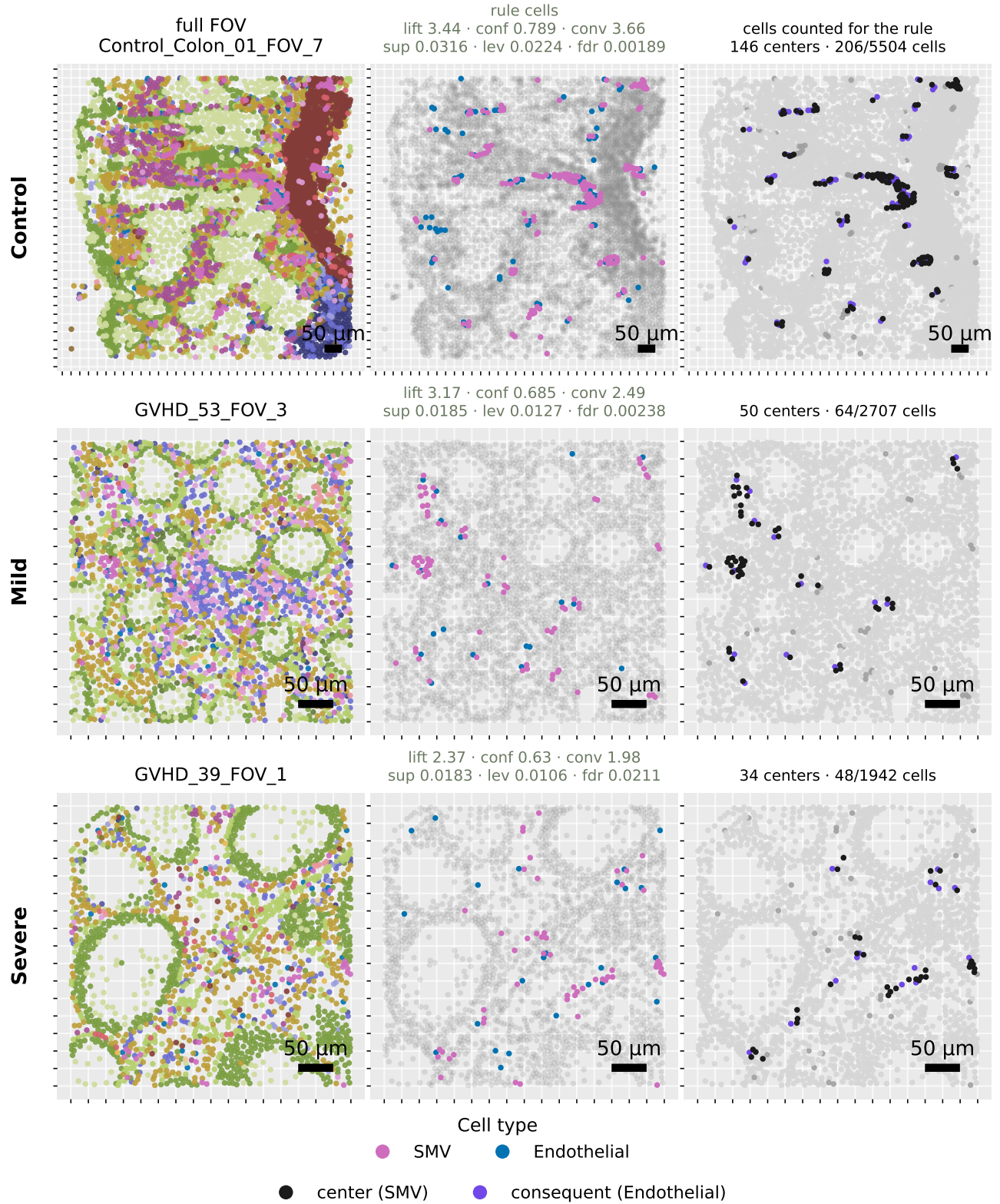


Figure 7: Each column shows a clinical stage, with the full FOV above and the same FOV highlighting only the two rule types below. One rule state is shown per figure. Maps are illustrative fields; highlighted cells are not individual detected interactions. Molecular signaling direction cannot be inferred from these maps.

SMV → Endothelial also recurs across 27 duodenal patients, extending the anatomical check beyond colon.

Duodenum · SMV → Endothelial

All = pooled organ · Clinical score · ≥ 20 cells/type

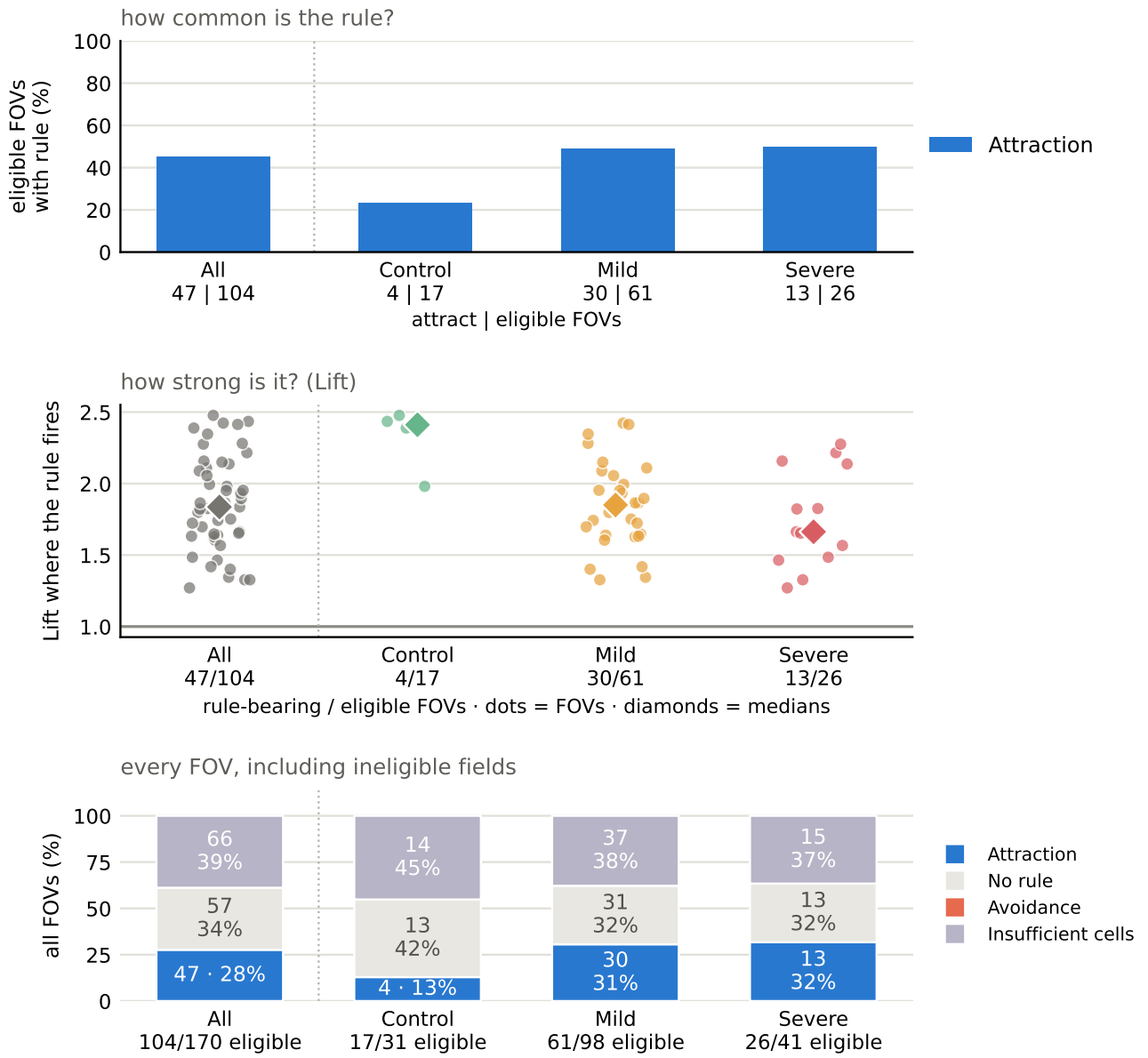


Figure 8: Calculations follow Figure 6. Attraction occurs in 47/104 eligible FOVs and 27/42 eligible patients, versus 30/104 fields and 17/42 patients in reverse. The recurrence across organs and patients makes this useful as a method check; the patient-level arrow comparison has $q = 0.080$. Endothelial-to-mural developmental signaling [8] does not predict the stronger spatial arrow.

The duodenal highlights show the same vessel-wall pairing in a second organ.

Duodenum · SMV → Endothelial

Attraction examples

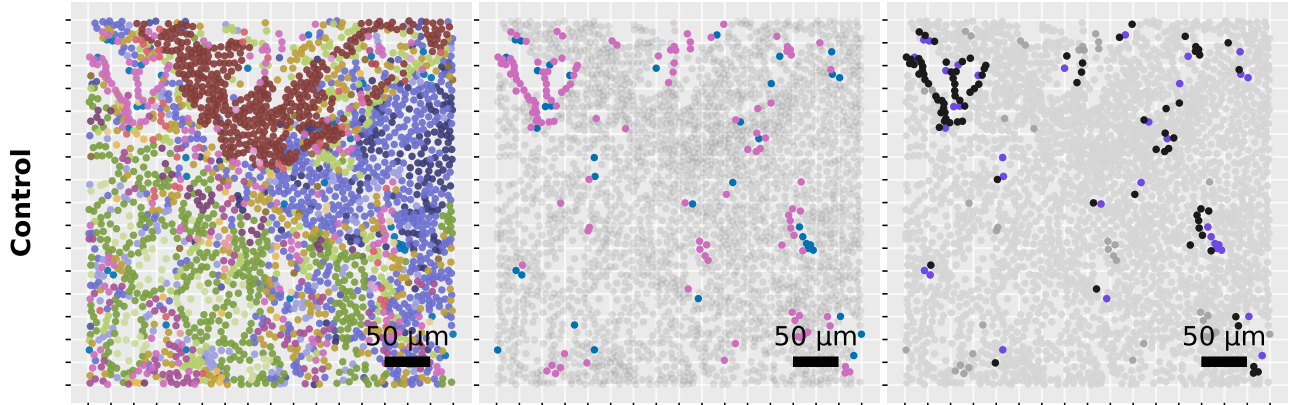
Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

rule cells

full FOV
Control_01_FOV_1

lift 2.39 · conf 0.697 · conv 2.34
sup 0.034 · lev 0.0198 · fdr 0.0166

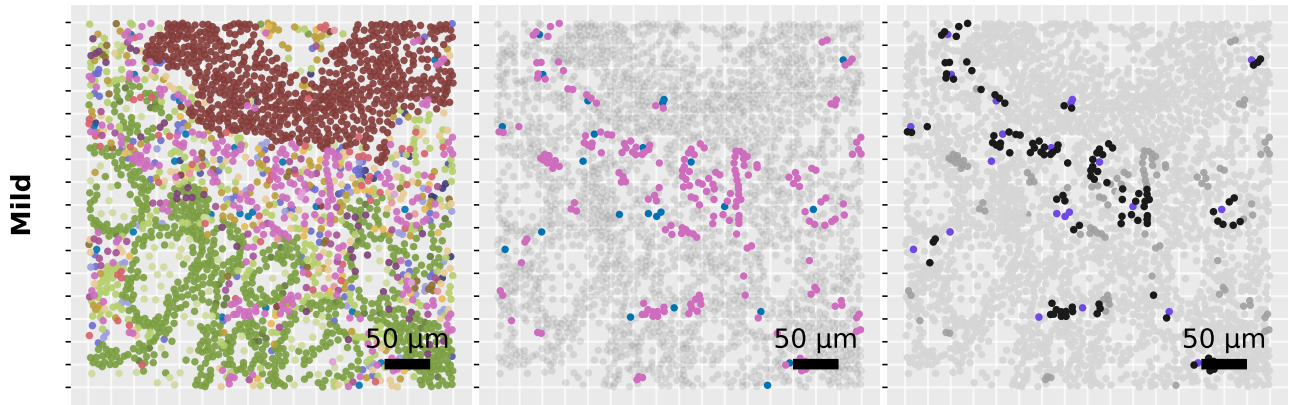
cells counted for the rule
85 centers · 119/2618 cells



GVHD_18_FOV_3

lift 1.86 · conf 0.519 · conv 1.5
sup 0.0526 · lev 0.0244 · fdr 0.00535

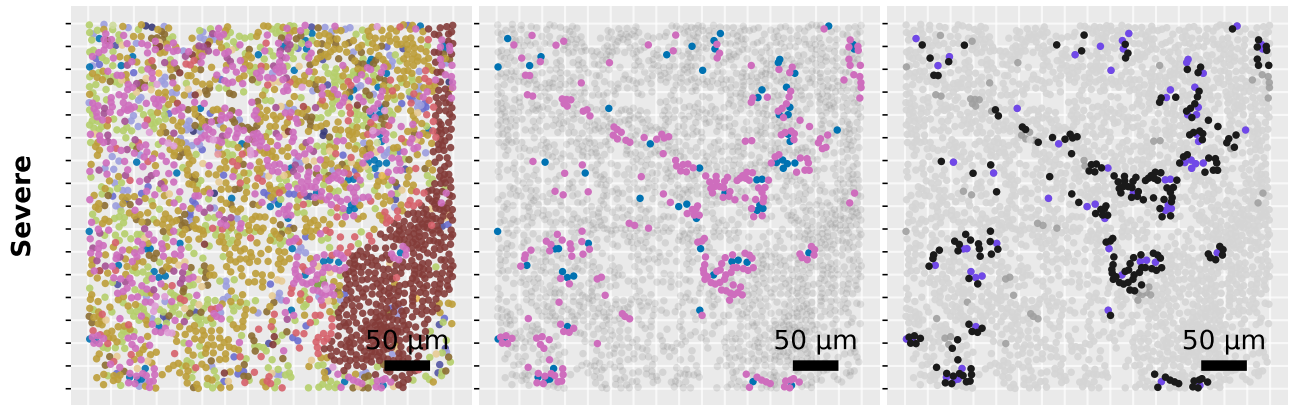
108 centers · 130/2525 cells



GVHD_26_FOV_1

lift 1.66 · conf 0.795 · conv 2.54
sup 0.0812 · lev 0.0324 · fdr 0.00175

205 centers · 272/2686 cells



Cell type

- SMV
- Endothelial
- center (SMV)
- consequent (Endothelial)

Figure 9: Maps and selection follow Figure 7.

Paneth → Epithelial is common in eligible fields, consistent with a restricted epithelial niche [9].

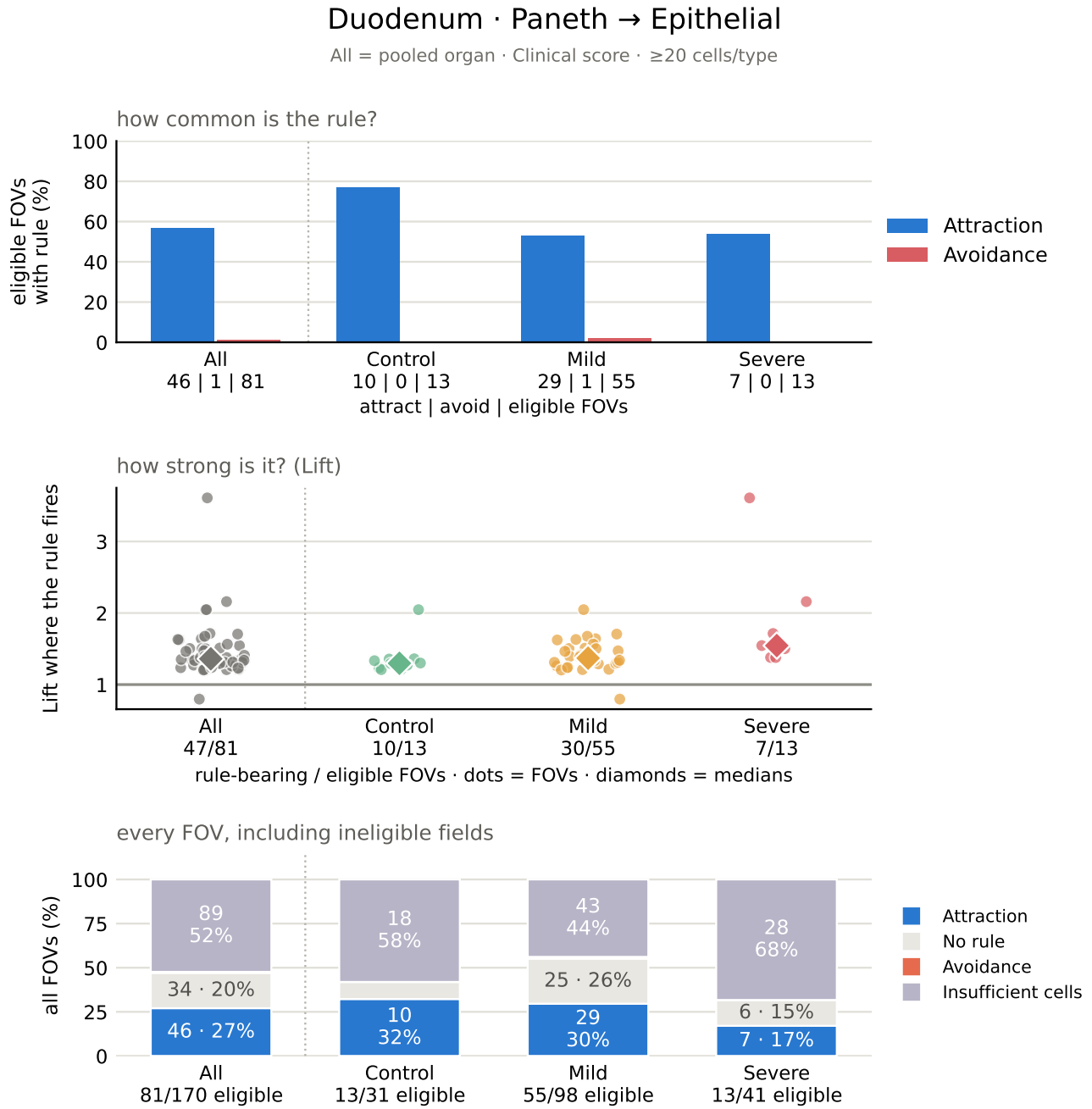


Figure 10: Calculations follow Figure 6. Attraction occurs in 46/81 fields versus 8/81 in reverse, across 28/36 versus 7/36 eligible patients (patient comparison $q < 0.001$). This recovers an asymmetric spatial rule, not Paneth-to-stem-cell signaling: the broad Epithelial label does not identify stem cells. The separate all-center benchmark in the notebook gives little positive Paneth-centered excess; its null and sampling differ from mining.

The highlights place Paneth cells among epithelial structures in typical attractive fields.

Duodenum · Paneth → Epithelial

Attraction examples

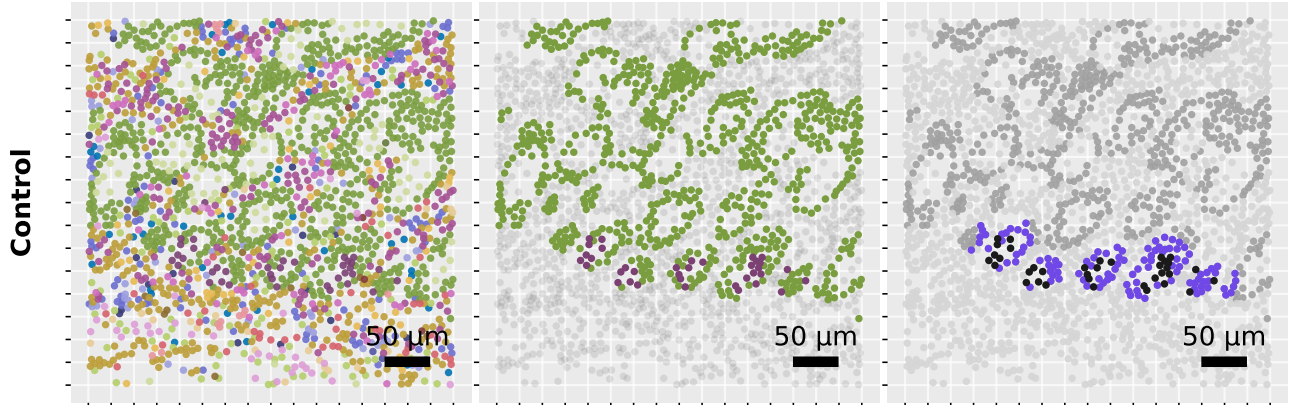
Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

rule cells

full FOV
Control_07_FOV_3

lift 1.3 · conf 1 · conv inf
sup 0.0193 · lev 0.00446 · fdr 0.0094

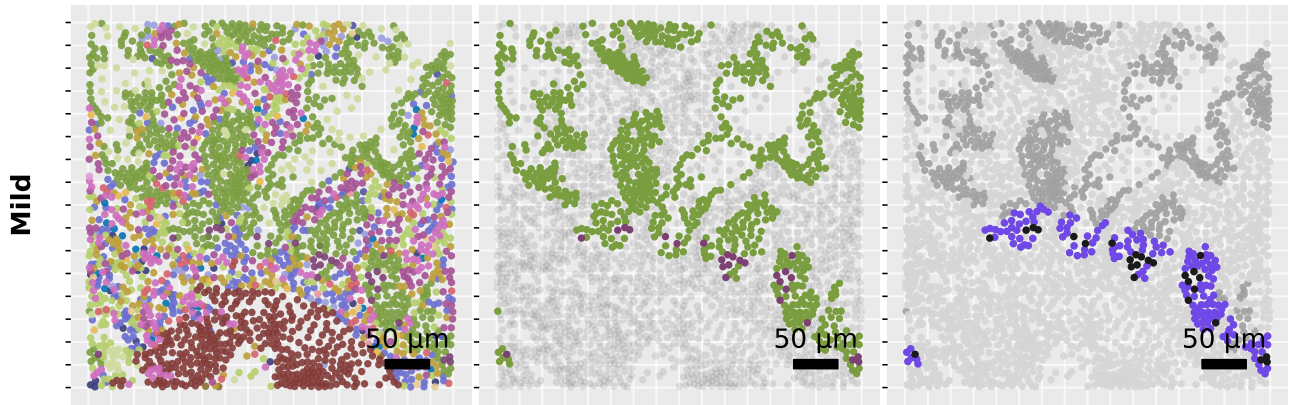
cells counted for the rule
36 centers · 144/1913 cells



GVHD_58_FOV_2

lift 1.39 · conf 1 · conv inf
sup 0.0111 · lev 0.00311 · fdr 0.00353

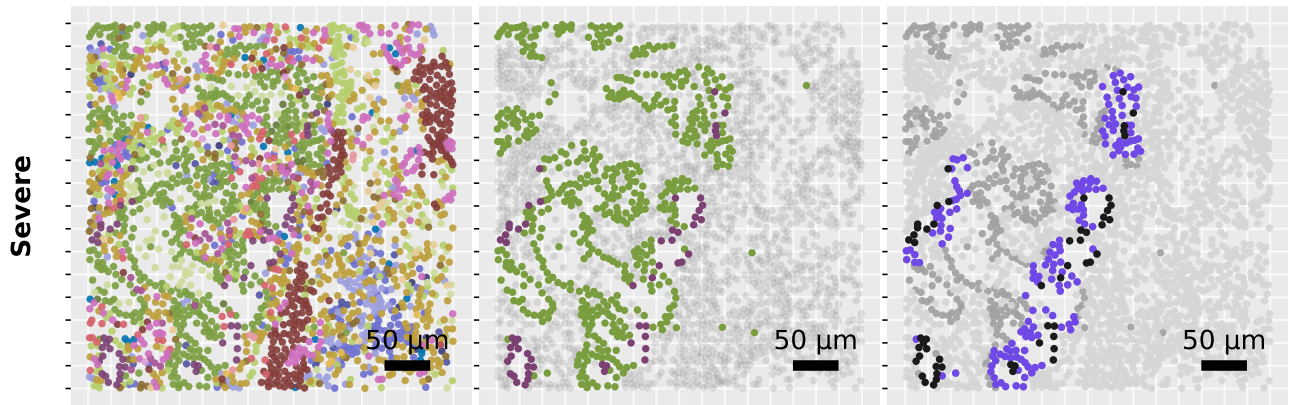
27 centers · 184/2628 cells



GVHD_19_FOV_3

lift 1.54 · conf 0.966 · conv 10.9
sup 0.0231 · lev 0.00816 · fdr 0.00465

56 centers · 226/2449 cells



Cell type

● Paneth ● Epithelial

● center (Paneth) ● consequent (Epithelial)

Figure 11: Maps and selection follow Figure 7.

Neuron → Macrophage recurs in colon fields, linking the mined rule to a known neuroimmune partnership [3].

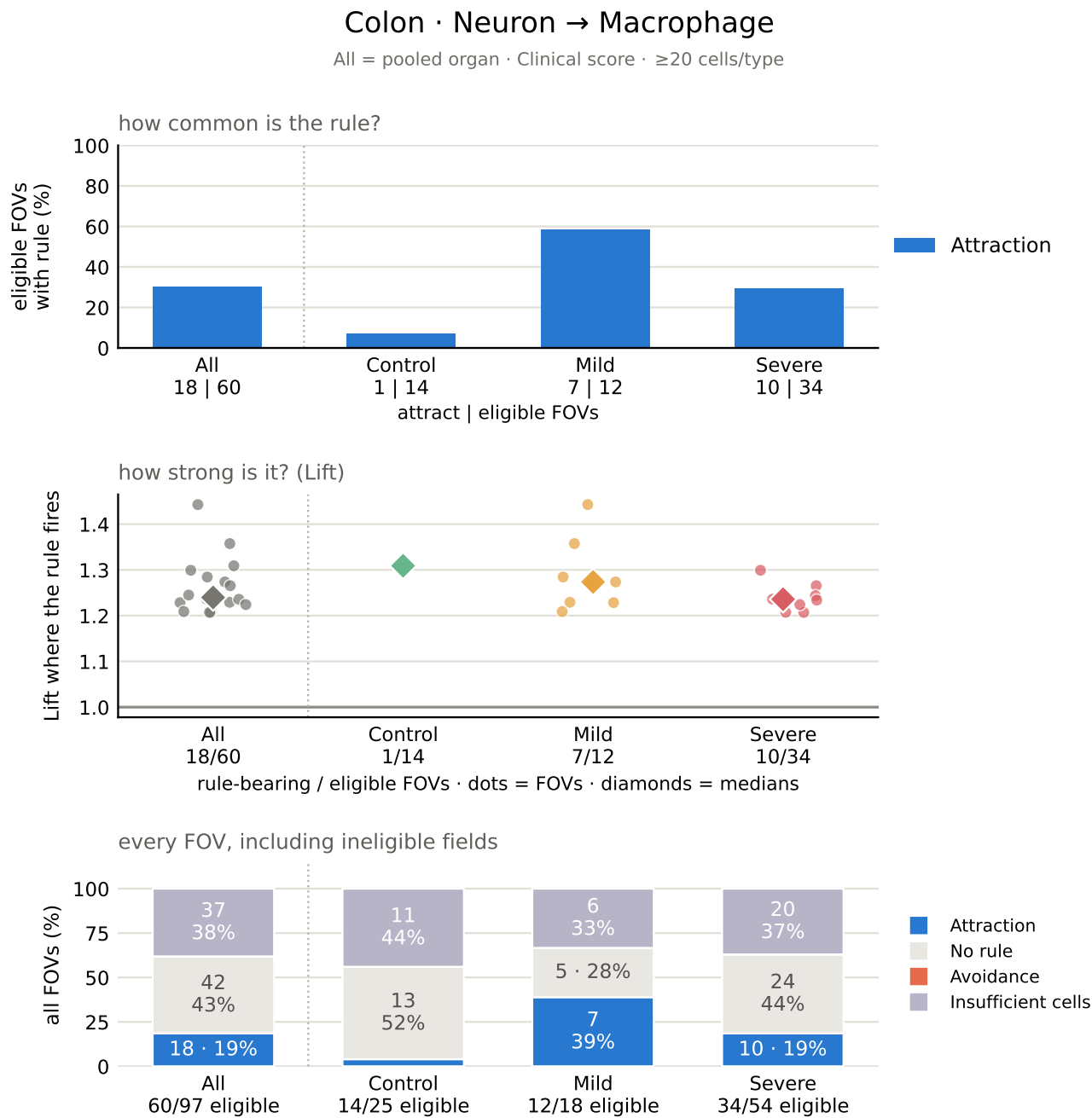


Figure 12: Calculations follow Figure 6. Attraction occurs in 18/60 fields versus 6/60 in reverse (patient comparison $q = 0.0053$). The reference describes reciprocal BMP2/CSF1 signaling; these broad labels and spatial rules cannot identify its molecular direction. The notebook's separate spatial-excess check favors the other arrow, reinforcing that distinction.

The highlighted neuron and macrophage cells give a tissue view of this asymmetric rule.

Colon · Neuron → Macrophage

Attraction examples

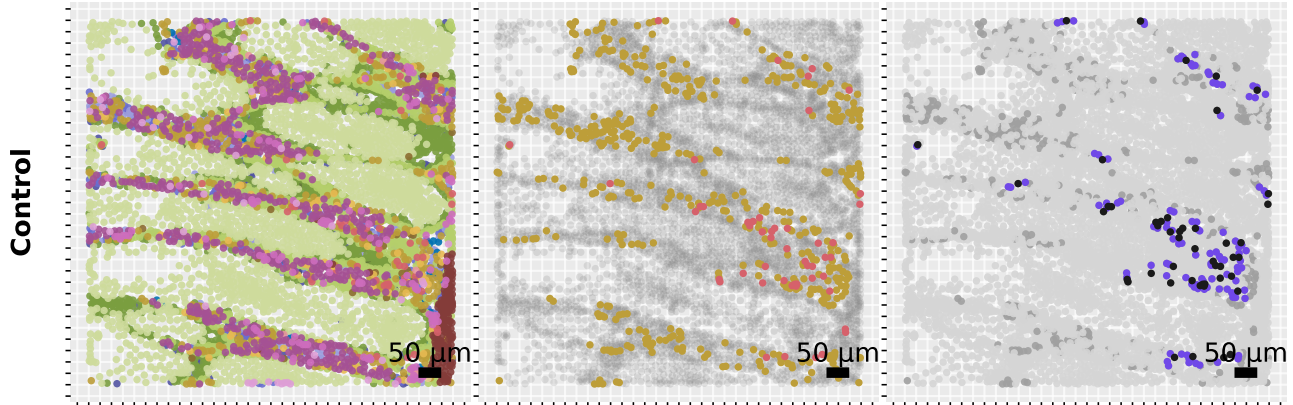
Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

rule cells

full FOV
Control_Colon_03_FOV_6

lift 1.31 · conf 0.941 · conv 4.78
sup 0.00958 · lev 0.00226 · fdr 0.00735

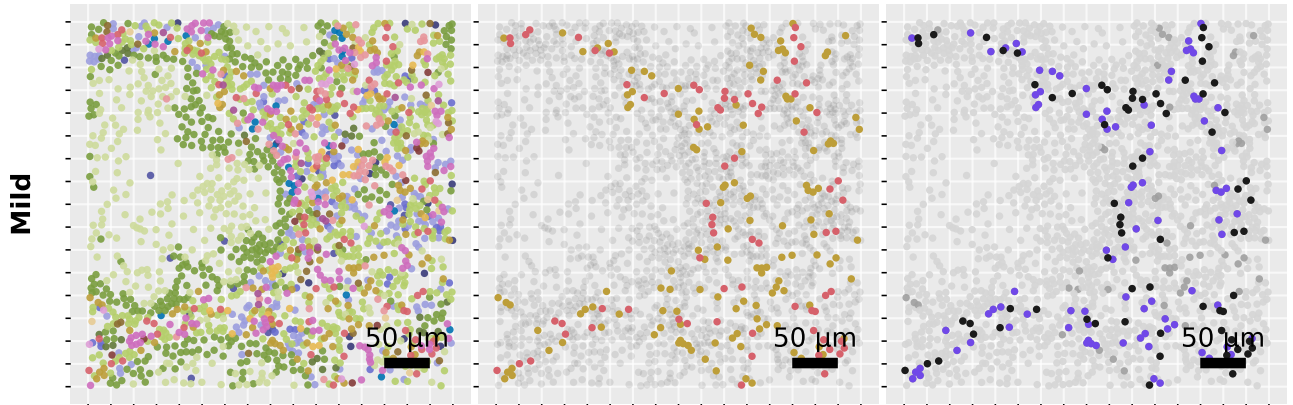
cells counted for the rule
48 centers · 159/6036 cells



GVHD_21_FOV_4

lift 1.27 · conf 0.947 · conv 4.87
sup 0.0388 · lev 0.00833 · fdr 0.00706

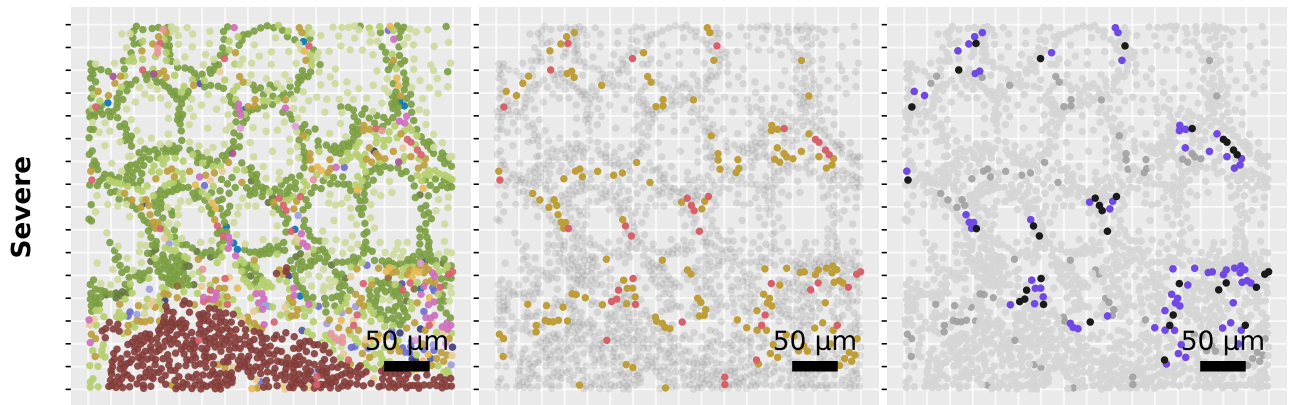
72 centers · 162/1956 cells



GVHD_50_FOV_3

lift 1.24 · conf 0.946 · conv 4.34
sup 0.019 · lev 0.00362 · fdr 0.0273

35 centers · 107/2088 cells



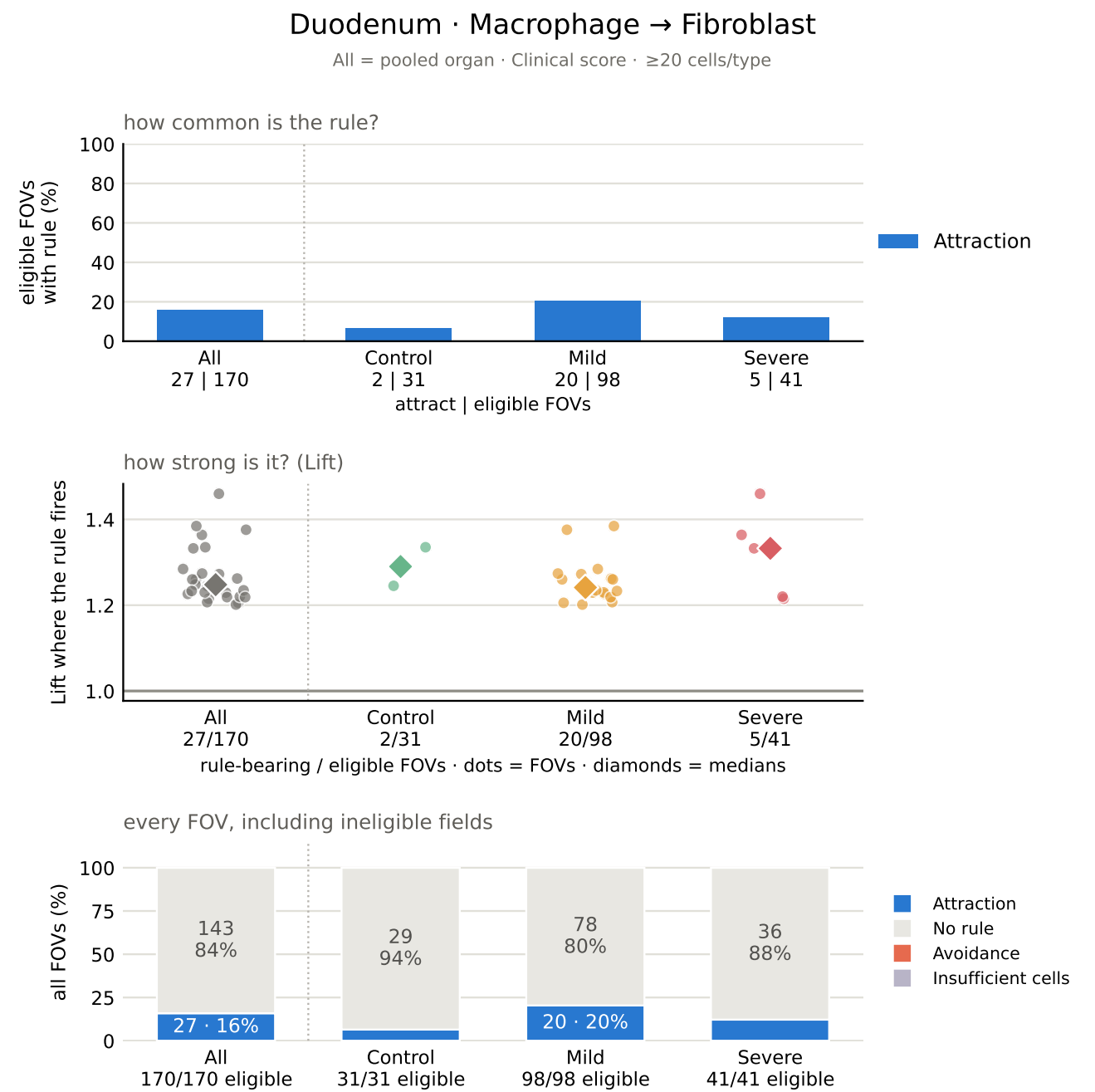
Cell type

● Neuron ● Macrophage

● center (Neuron) ● consequent (Macrophage)

Figure 13: Maps and selection follow Figure 7.

Macrophage → Fibroblast attraction recurs in 19 duodenal patients, giving a localized example of a literature-motivated partnership [4].



The highlights locate macrophages and fibroblasts in fields behind this minority rule.

Duodenum · Macrophage → Fibroblast

Attraction examples

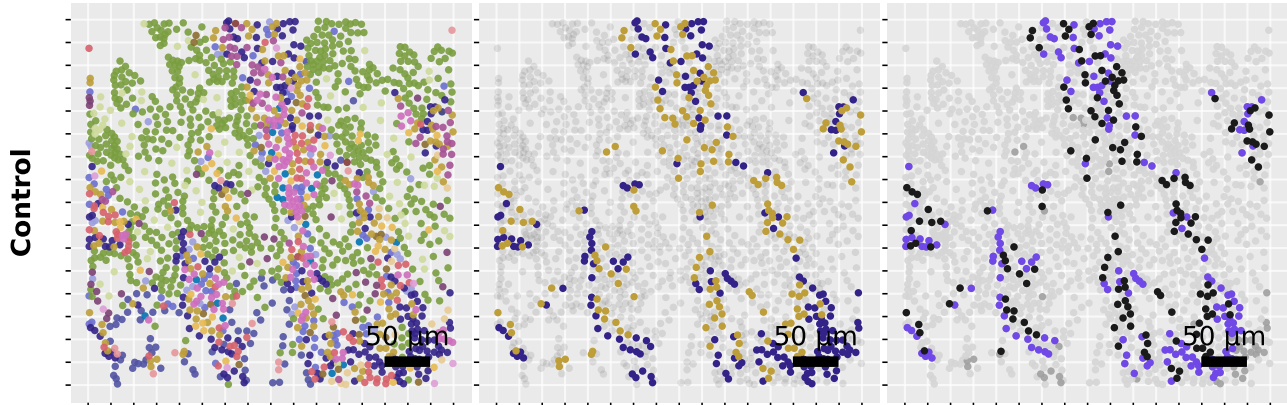
Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

rule cells

full FOV
Control_06_FOV_1

lift 1.34 · conf 0.914 · conv 3.66
sup 0.0997 · lev 0.025 · fdr 0.0133

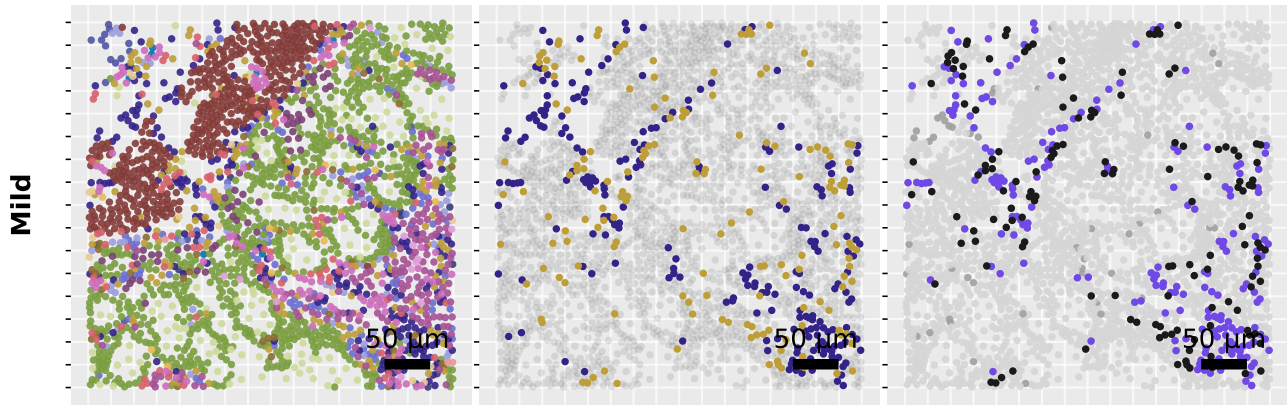
cells counted for the rule
159 centers · 330/1724 cells



GVHD_49_FOV_1

lift 1.25 · conf 0.887 · conv 2.57
sup 0.0602 · lev 0.012 · fdr 0.0041

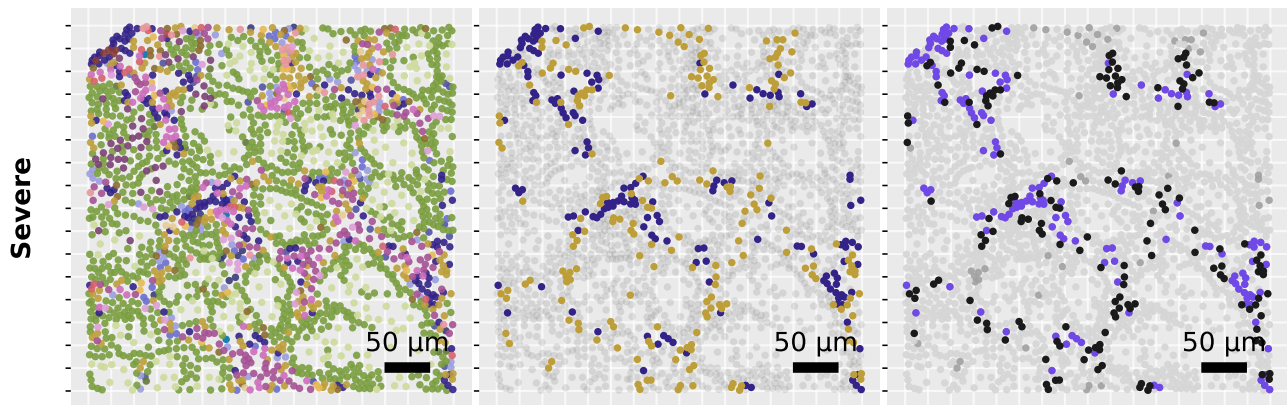
142 centers · 337/2531 cells



GVHD_57_FOV_2

lift 1.33 · conf 0.801 · conv 2
sup 0.076 · lev 0.019 · fdr 0.00694

165 centers · 323/2178 cells



Cell type

● Macrophage ● Fibroblast

● center (Macrophage) ● consequent (Fibroblast)

Figure 15: Maps and selection follow Figure 7. Typical attracting fields are shown, not the many eligible fields without a passing rule.

2 Paper alignment

These rules were chosen from claims in the spatial atlas [1] before checking our rule rankings. Stage comparisons here use biopsy-level tests where available.

The atlas describes altered crypt architecture; the preselected Epithelial → Goblet rule remains common in eligible duodenal fields [1].

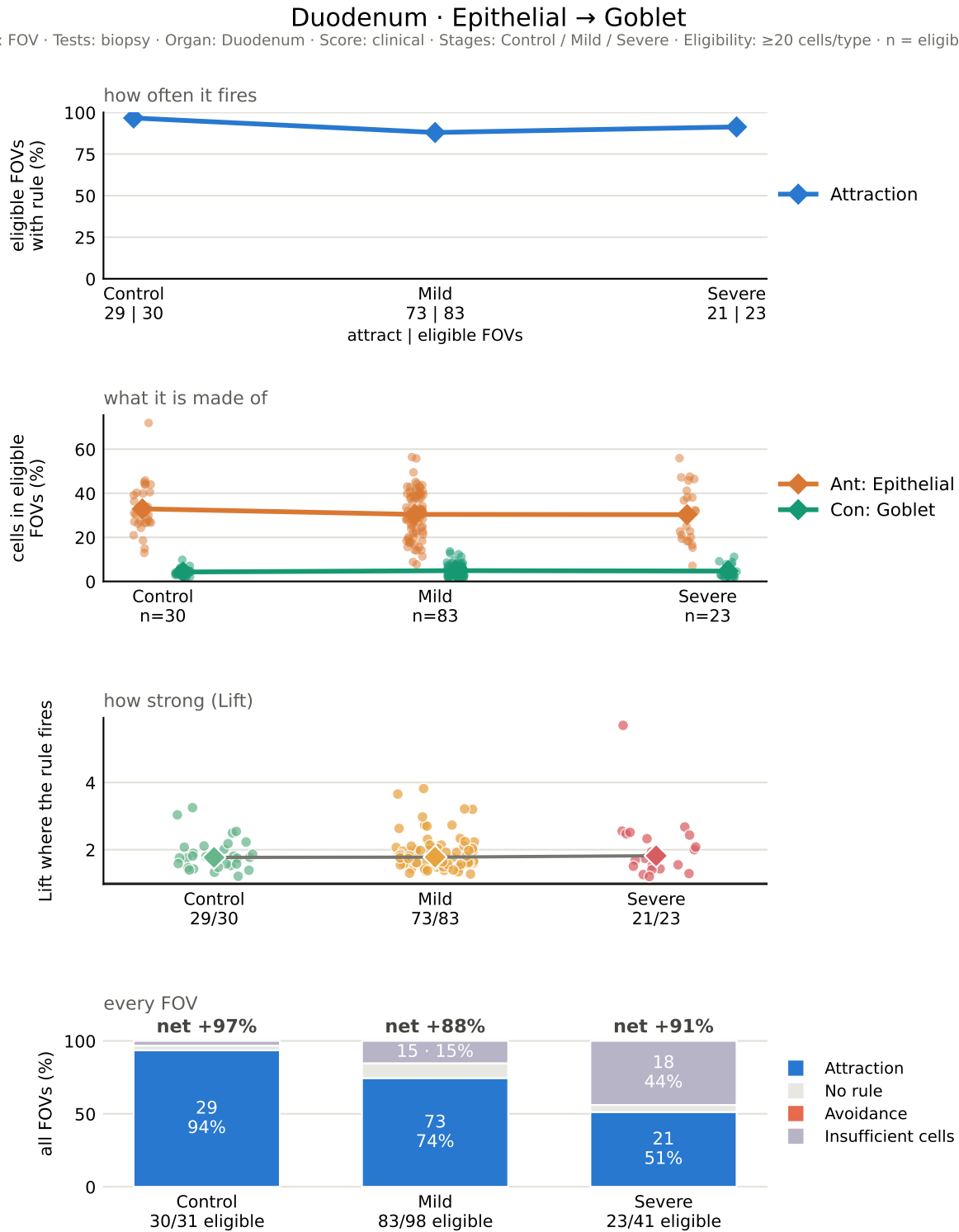


Figure 16: Read as in Figure 3. Attraction appears in 29/30 Control and 21/23 Severe eligible FOVs.

The highlighted cells place this common rule along epithelial and goblet-cell structures.

Duodenum · Epithelial → Goblet

Attraction examples

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

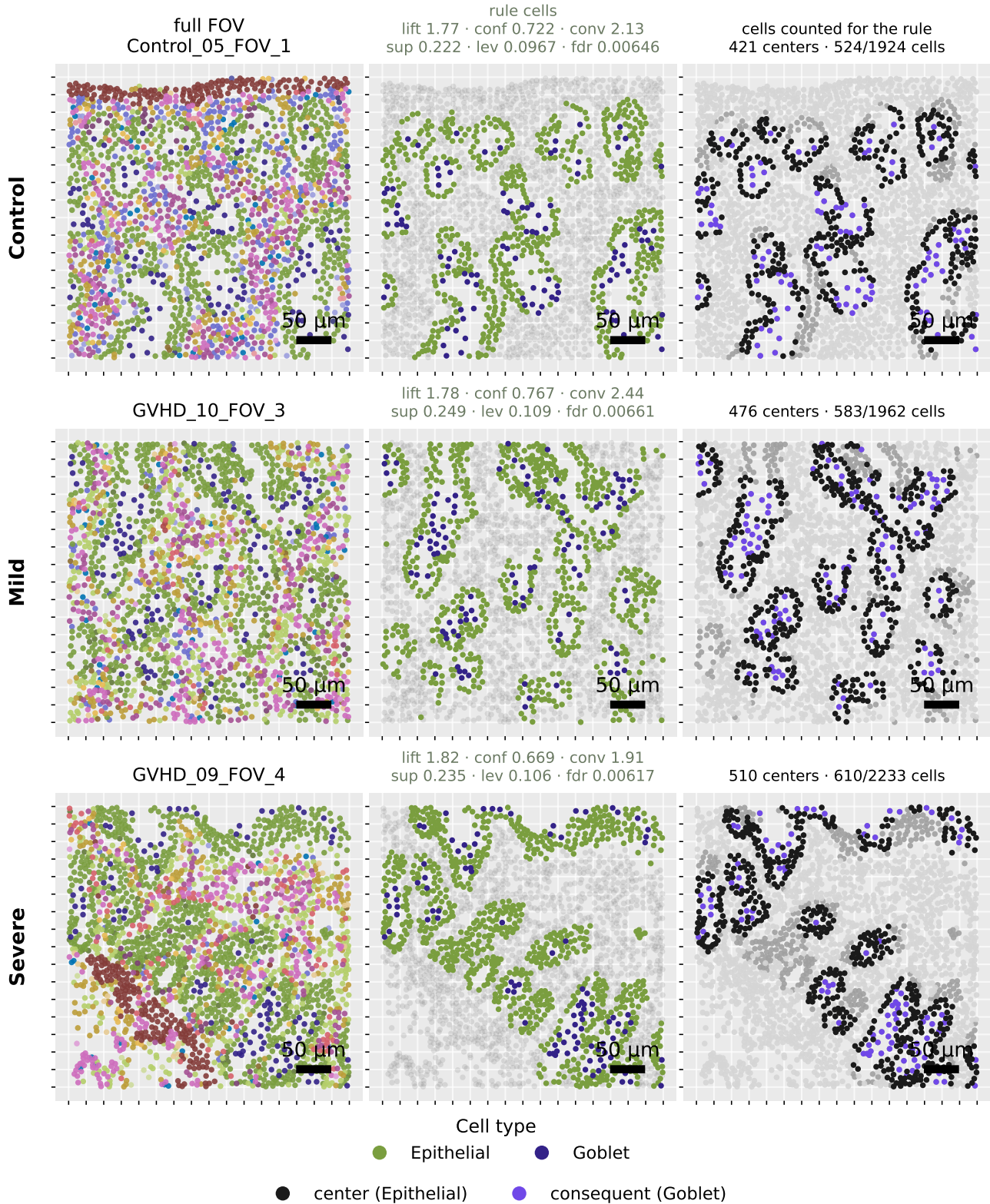


Figure 17: Columns are stages or time windows, one rule state per figure. Each example appears first as a full FOV and then with only the rule's cell types highlighted. Rule-bearing examples are nearest their state's median Lift; a no-rule figure uses eligible comparison fields.

The eligible no-rule fields show how the same cell types appear when this rule is not mined.

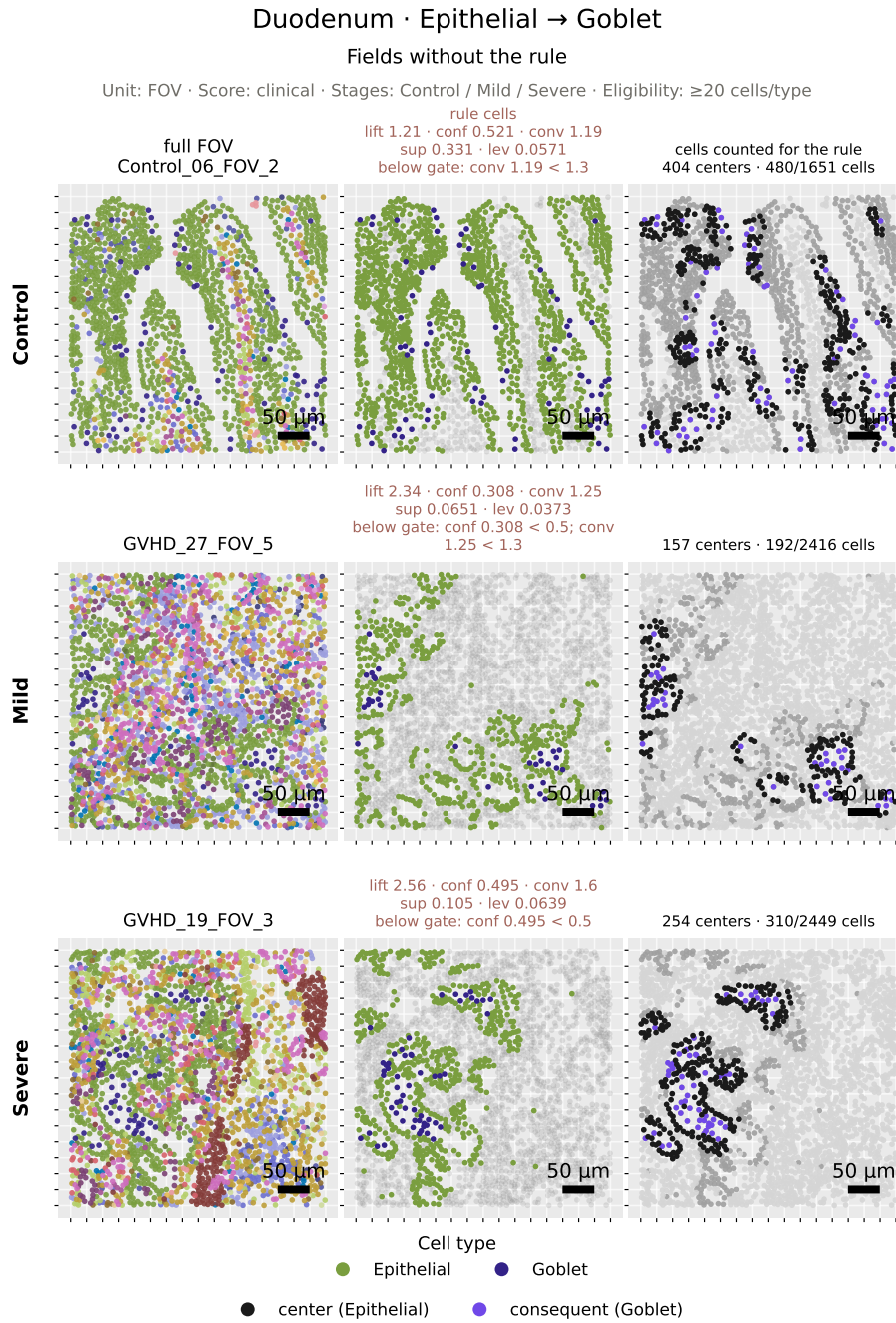


Figure 18: Maps and example selection follow Figure 17.

The reverse Goblet → Epithelial rule is less common, showing that the two conditional arrows describe different patterns in the same fields.

Duodenum · Goblet → Epithelial

Plots: FOV · Tests: biopsy · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs

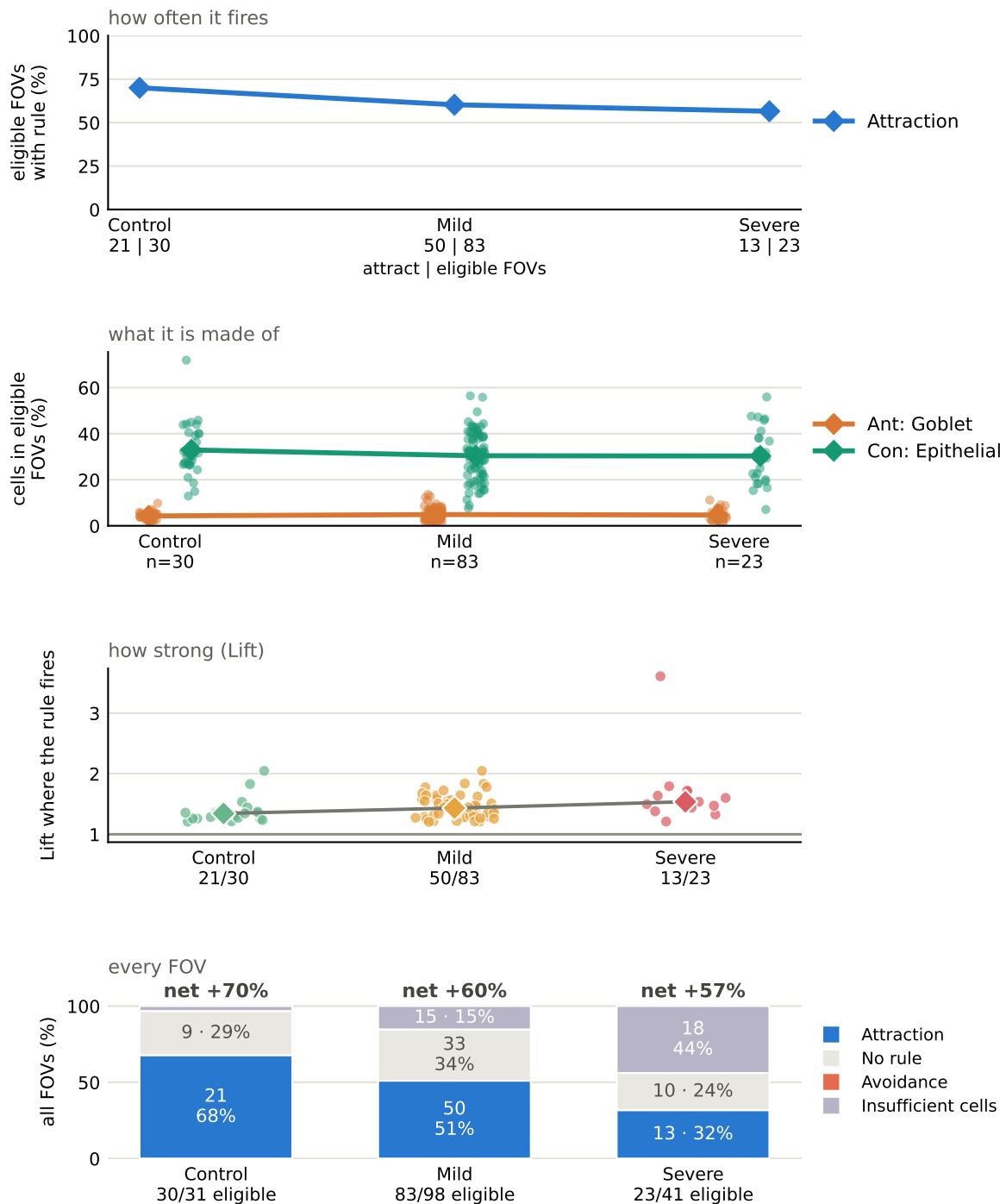


Figure 19: Read as in Figure 3. Attraction appears in 21/30 Control and 13/23 Severe eligible fields, compared with 29/30 and 21/23 in the forward arrow. The separate biopsy-level ordered Lift test gives FDR $q = 0.032$; this direction was preselected from the atlas pair.

The atlas reports Paneth-cell depletion and crypt injury; among eligible surviving fields, Paneth → Epithelial stays attractive and its Lift rises across stages [1, 9].

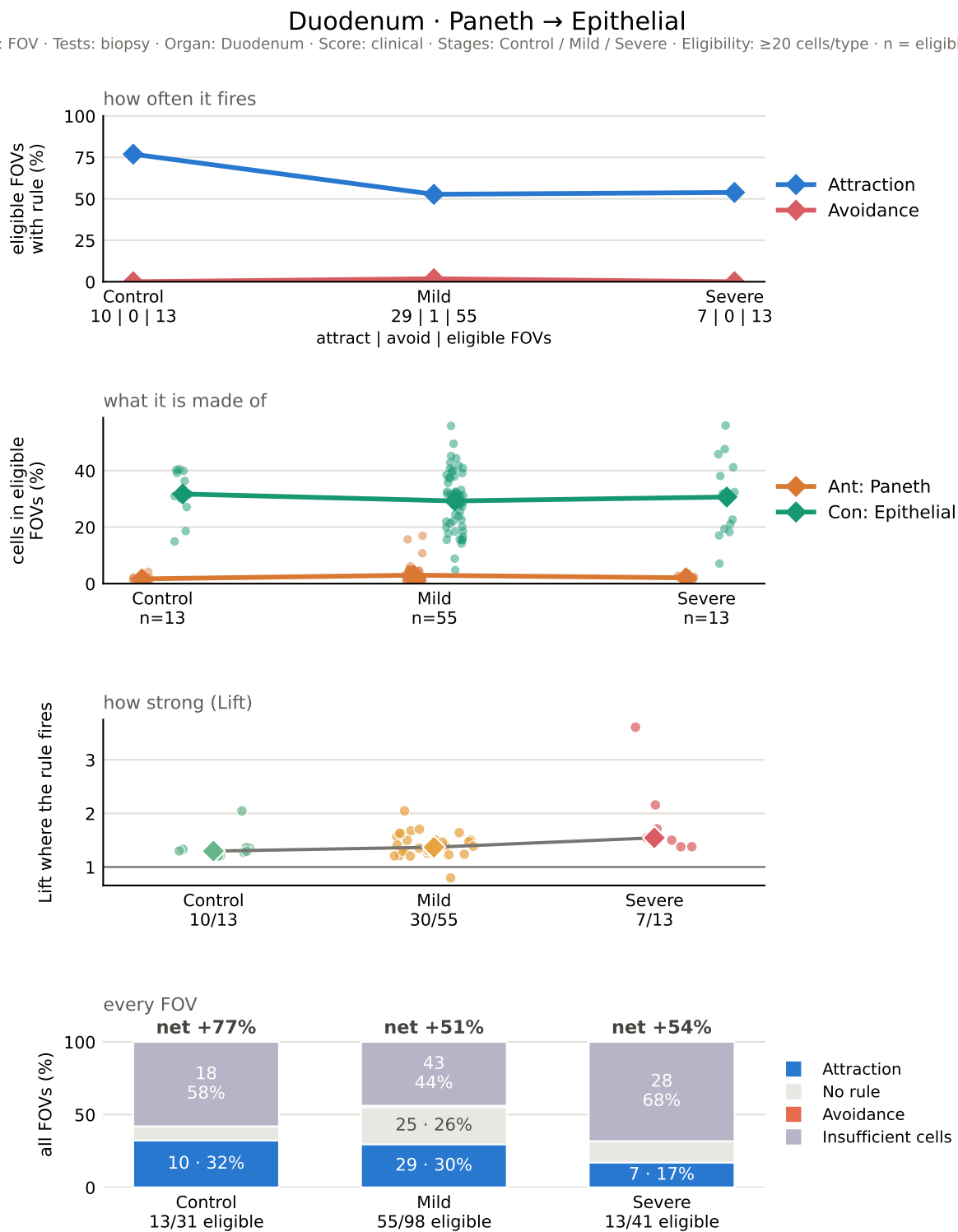


Figure 20: Read as in Figure 3. The ordered biopsy-level attraction-Lift test gives FDR $q = 0.0196$ in the separate gap screen. Only 13 FOVs are eligible in each endpoint stage, so the apparent prevalence drop is exploratory.

The examples place Paneth cells beside epithelial structures, making the mined relationship visible.

Duodenum · Paneth → Epithelial

Attraction examples

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

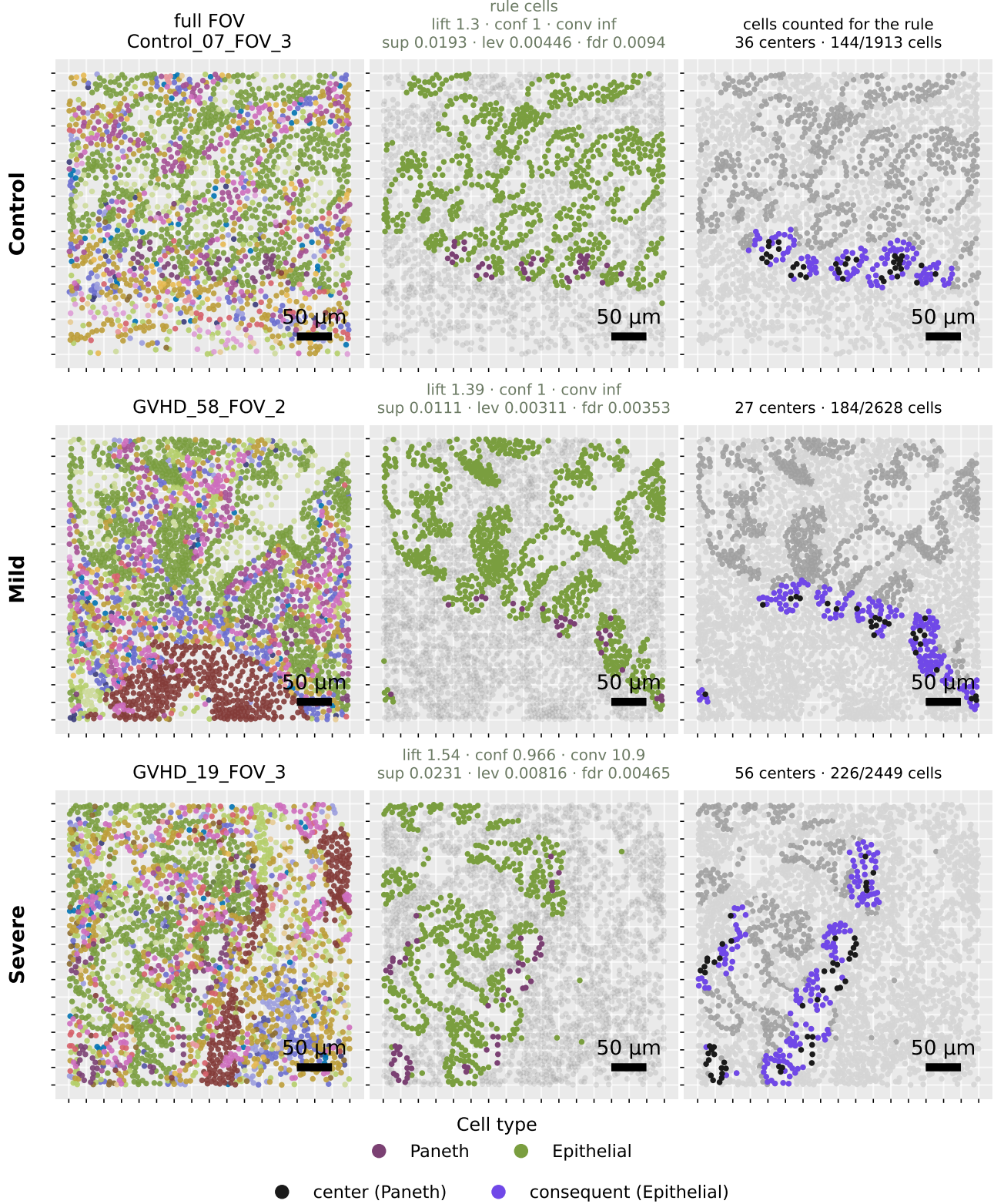


Figure 21: Maps and example selection follow Figure 17.

The atlas reports altered crypts and CD8 T-cell enrichment in some patients; the preselected Goblet → CD8T avoidance rises from 13/28 to 14/17 eligible duodenal fields [1].

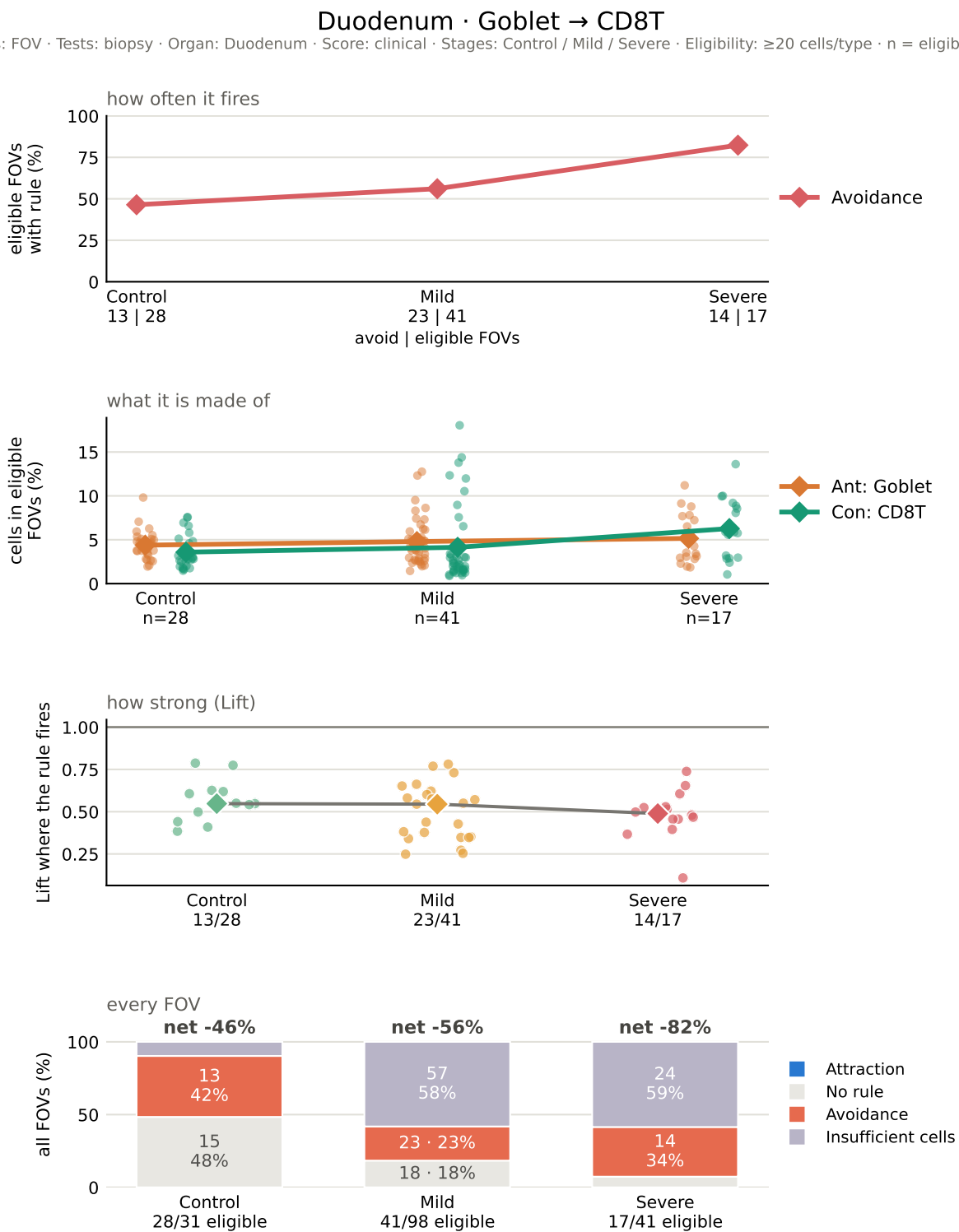


Figure 22: Read as in Figure 3. This is an exploratory stage difference; 17 Severe FOVs are eligible. The reverse CD8T → Goblet arrow also gains avoidance, from 6/28 to 9/17 fields.

The highlighted goblet and CD8 T cells show the separated arrangement behind the avoidance counts.

Duodenum · Goblet → CD8T

Avoidance examples

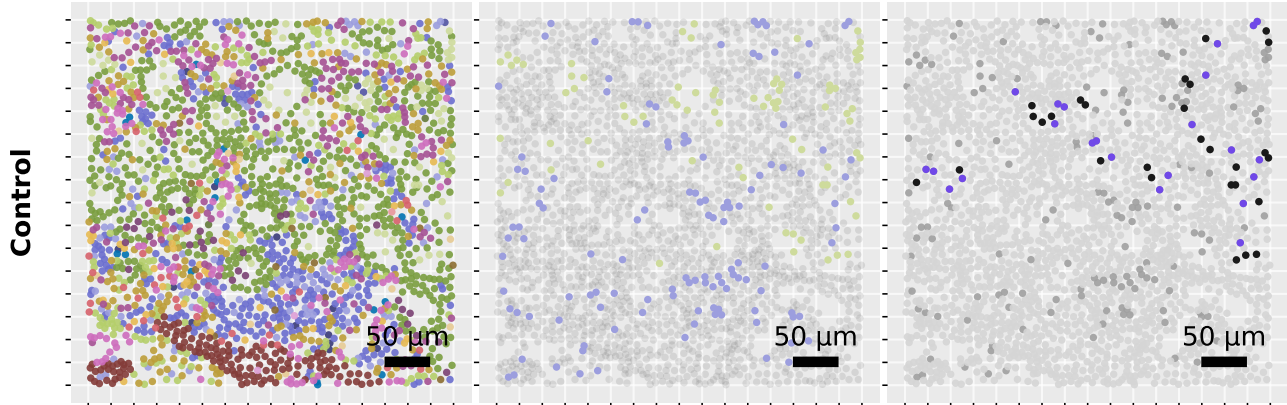
Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

rule cells

full FOV
Control_01_FOV_3

lift 0.548 · conf 0.35 · conv 0.555
sup 0.0131 · lev -0.0108 · fdr 0.0039

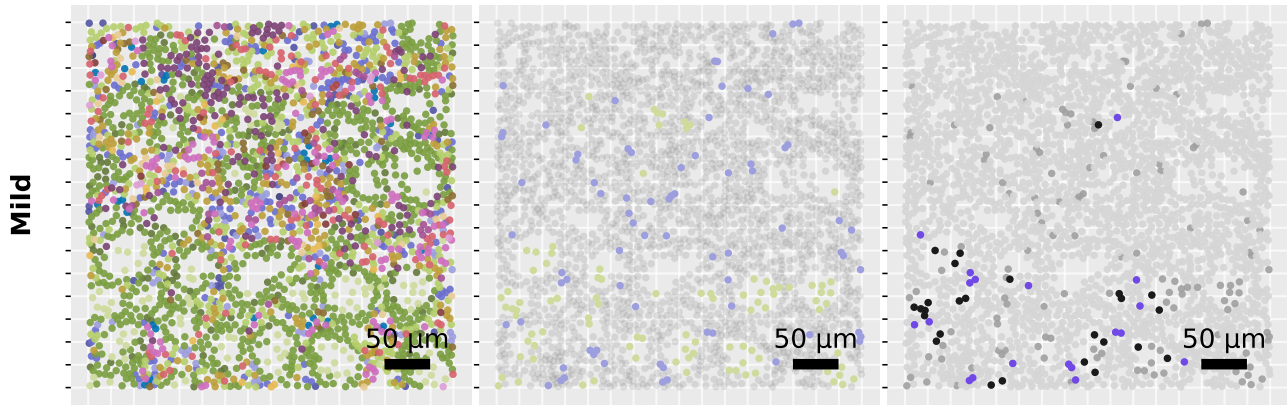
cells counted for the rule
28 centers · 51/2157 cells



GVHD_51_FOV_1

lift 0.545 · conf 0.277 · conv 0.68
sup 0.0103 · lev -0.00861 · fdr 0.00957

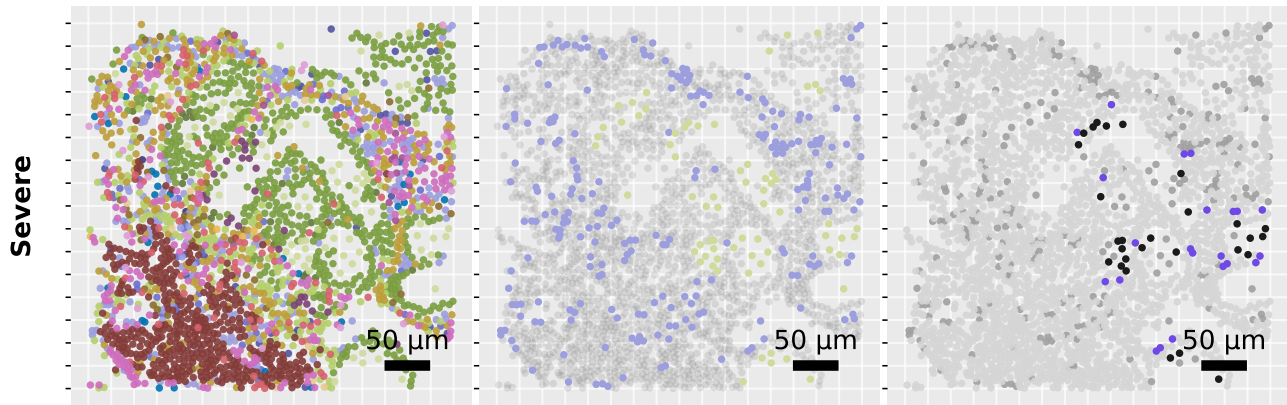
26 centers · 45/2527 cells



GVHD_14_FOV_5

lift 0.498 · conf 0.424 · conv 0.257
sup 0.0118 · lev -0.0119 · fdr 0.00357

28 centers · 51/2413 cells



Cell type

● Goblet ● CD8T

● center (Goblet) ● consequent (CD8T)

Figure 23: Maps and example selection follow Figure 17.

The atlas reports disrupted immune organization and fewer IgA plasma cells; the preselected colonic CD4T → Plasma attraction falls from 14/25 Control to 1/24 Severe eligible fields [1].

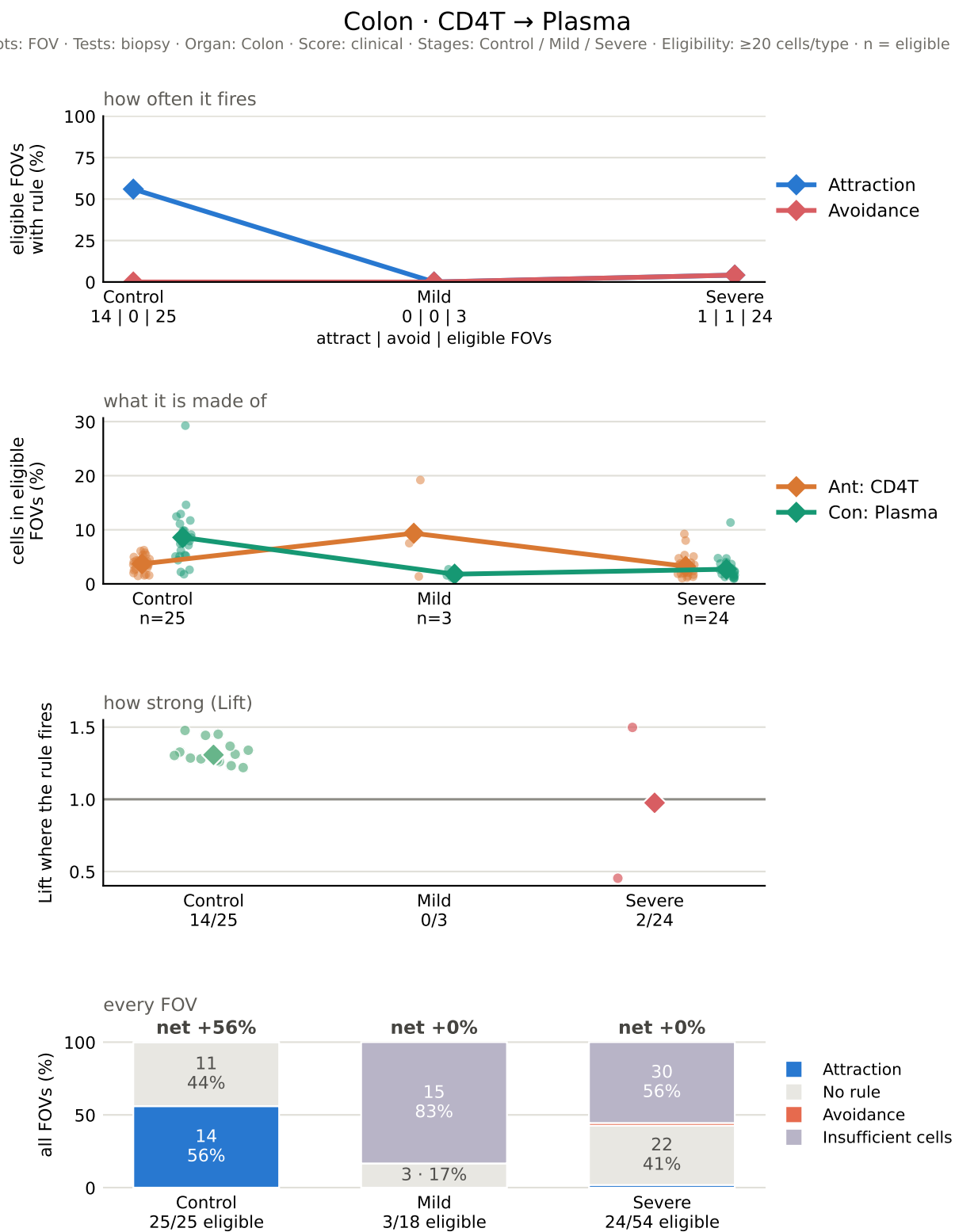


Figure 24: Read as in Figure 3. This atlas-selected relationship also entered the automatic visible-effect screen, but its biopsy-level FDR is $q = 0.323$. Only 3 Mild fields are eligible; the Control–Severe comparison is exploratory, and the abundance row shows fewer Plasma cells within eligible Severe FOVs.

The Control maps show the colonic CD4 T-cell and plasma-cell arrangement where attraction fires.

Colon · CD4T → Plasma

Attraction examples

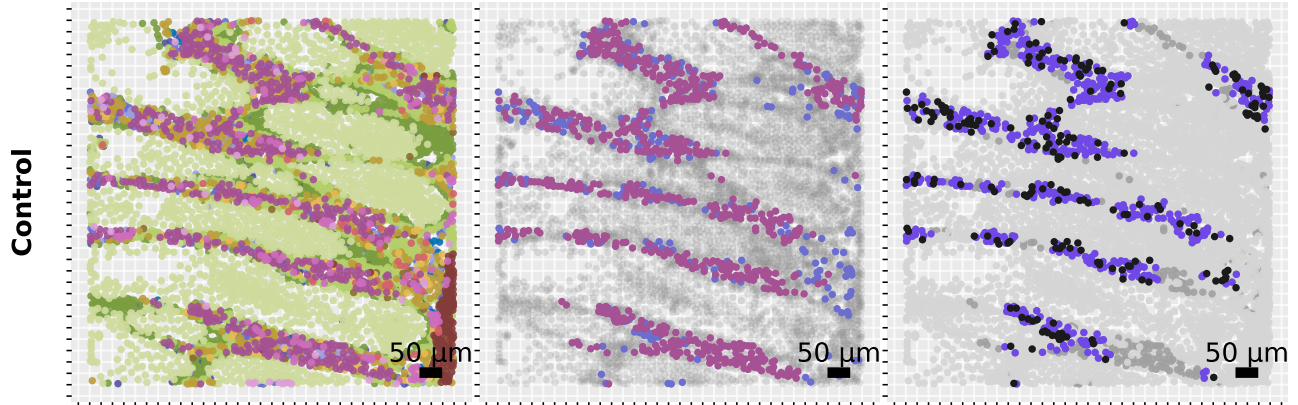
Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

rule cells

lift 1.3 · conf 0.83 · conv 2.14

sup 0.0429 · lev 0.00999 · fdr 0.00213

cells counted for the rule
215 centers · 706/6036 cells



Mild

No eligible attraction example

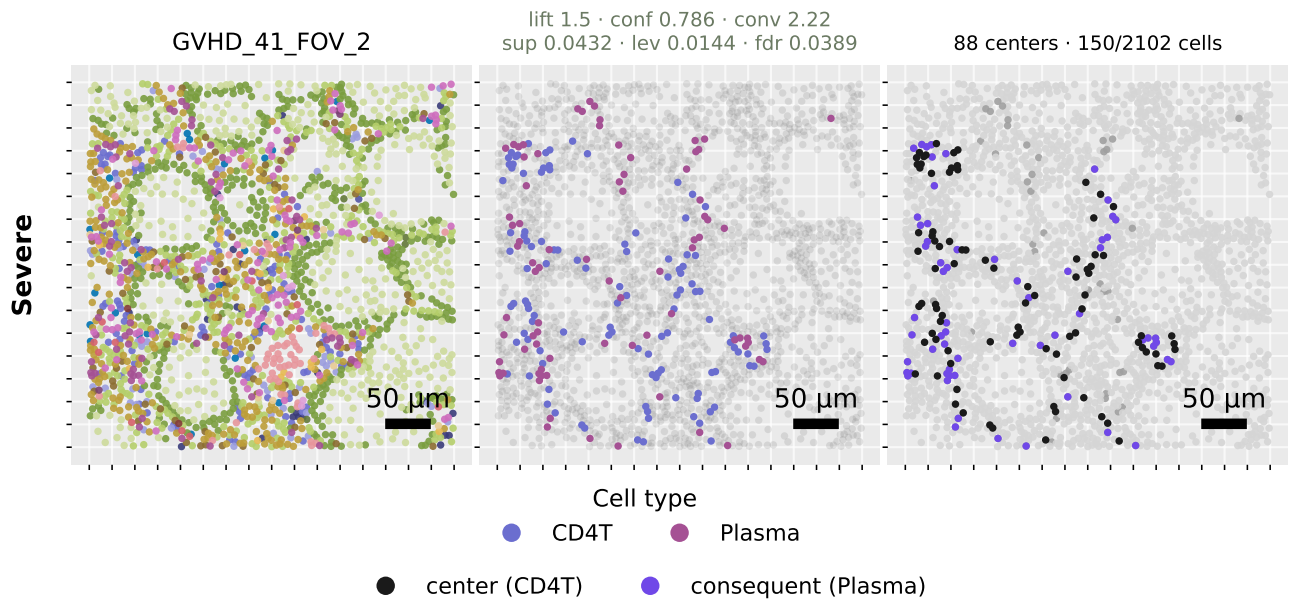


Figure 25: Maps and example selection follow Figure 17.

3 New findings

Automatic severity and time screens ranked both rule directions. The visually clear results below are exploratory FOV comparisons, without patient-level significance claims. The statistical-stream result, Paneth → Epithelial Lift, is in Paper alignment.

3.1 Goblet and muscle compartments become less separated in colon

Colon Goblet → Muscle avoidance falls from 19/20 Control to 13/23 Severe eligible fields, and the reverse arrow changes in the same direction.

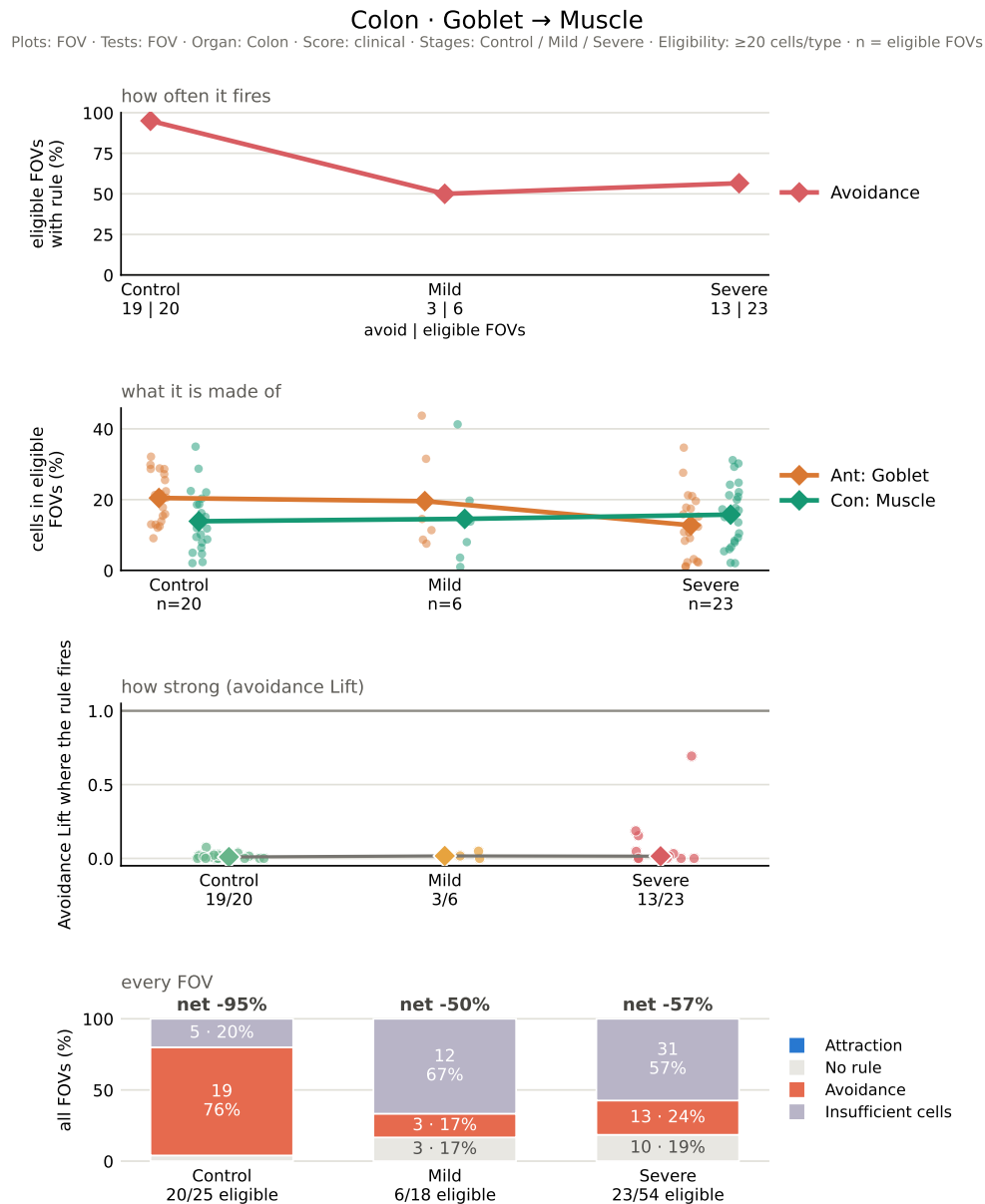


Figure 26: Read as in Figure 3. The reverse Muscle → Goblet avoidance falls from 20/20 to 15/23 eligible fields. The forward rule occurs in 4/4 eligible Control patients and 7/10 Severe patients. Four Control patients limit independent inference; the paired FOV pattern remains exploratory.

The highlighted fields show goblet cells and muscle where their avoidance rule is mined.

Colon · Goblet → Muscle

Avoidance examples

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

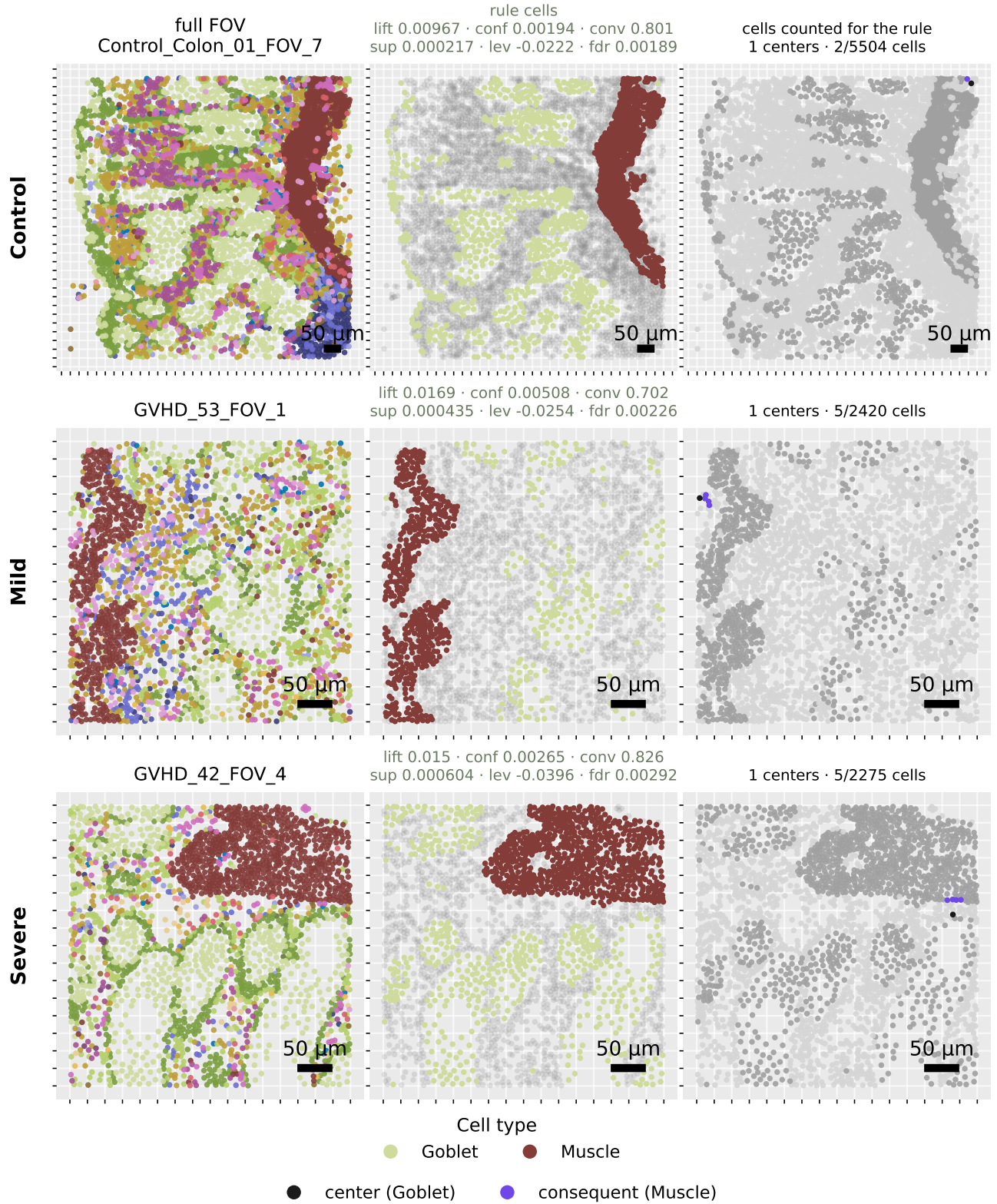


Figure 27: Maps and example selection follow Figure 17.

3.2 Immune cells move relative to the duodenal wall

CD4T → Goblet avoidance falls from 23/30 Control fields to 4/21 Severe fields, suggesting less separation between cells with known inflammatory interactions [5, 6].

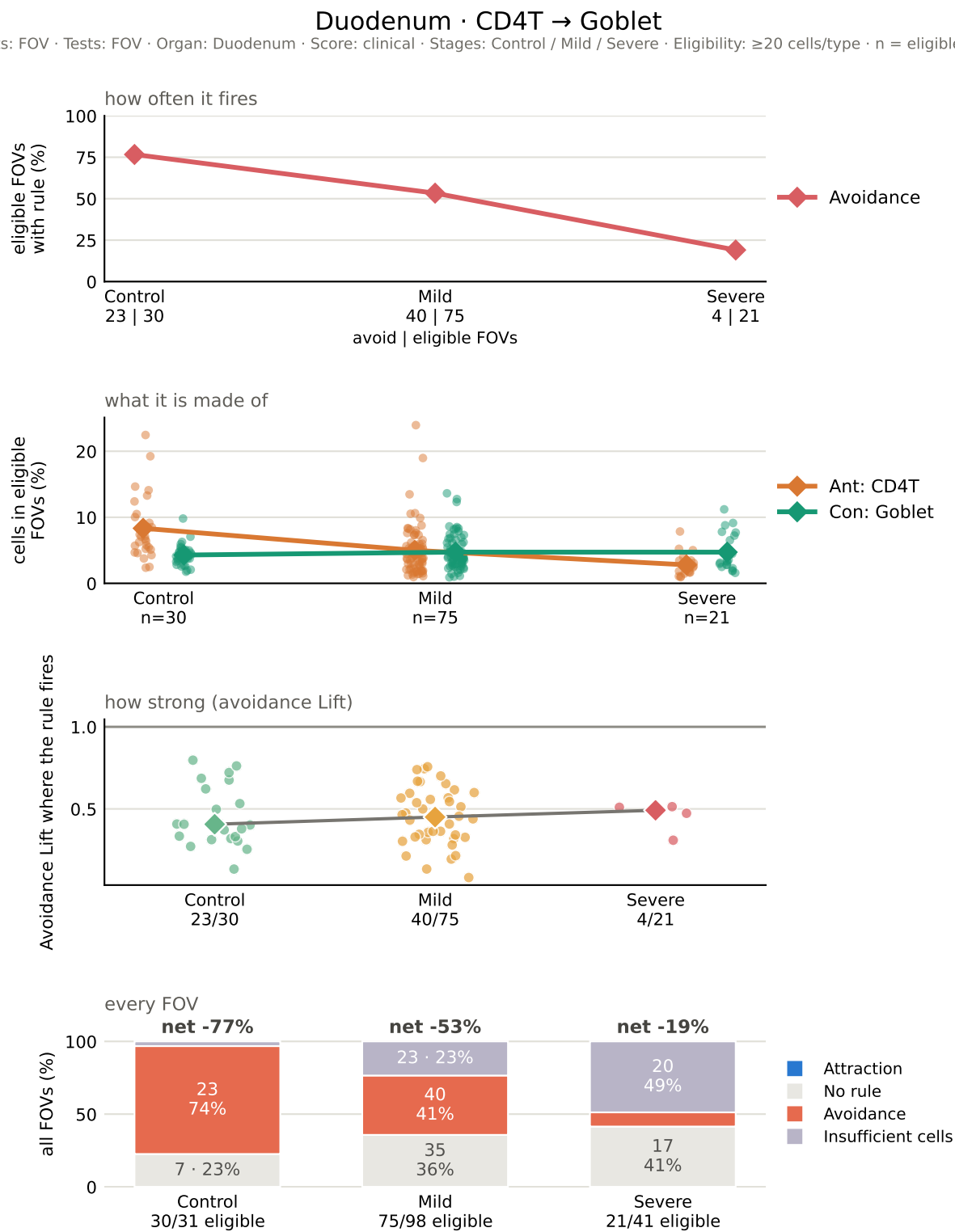


Figure 28: Read as in Figure 3. The Control–Severe avoidance gap is 58 percentage points. The rule occurs in 9/10 versus 4/7 eligible patients; these are descriptive counts, not a patient-level significance test.

The highlighted CD4 T and goblet cells show the separated arrangement seen in avoiding fields.

Duodenum · CD4T → Goblet

Avoidance examples

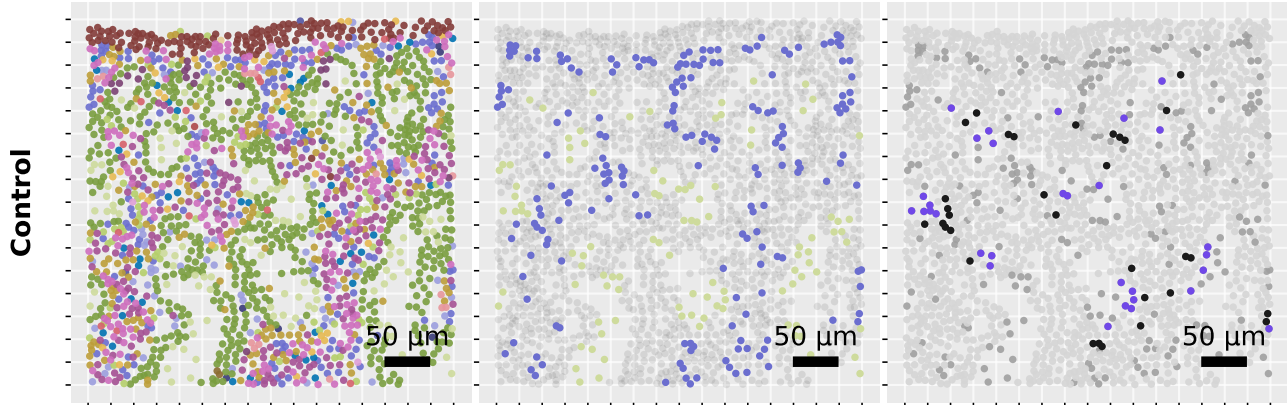
Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

rule cells

full FOV
Control_05_FOV_1

lift 0.406 · conf 0.166 · conv 0.71
sup 0.0169 · lev -0.0247 · fdr 0.00646

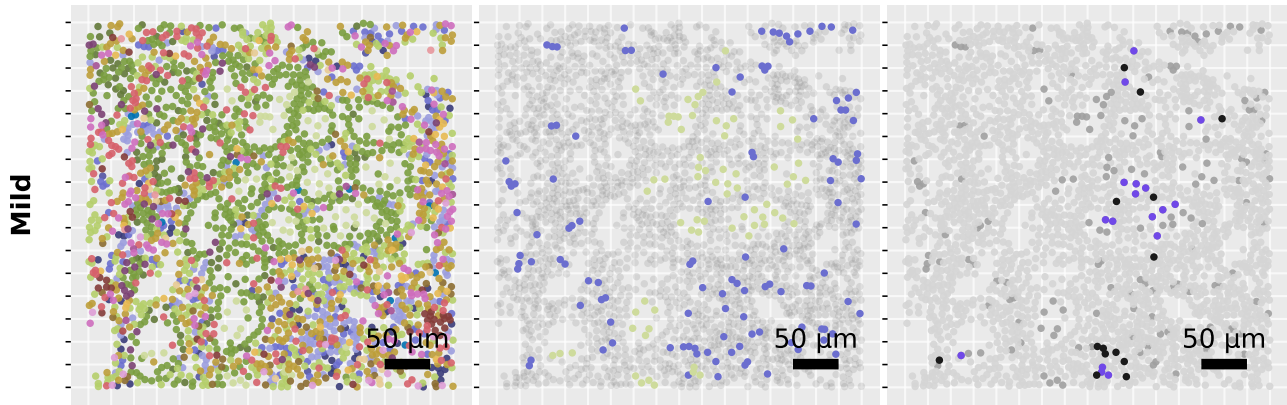
cells counted for the rule
32 centers · 63/1924 cells



GVHD_03_FOV_4

lift 0.445 · conf 0.121 · conv 0.829
sup 0.00573 · lev -0.00714 · fdr 0.00655

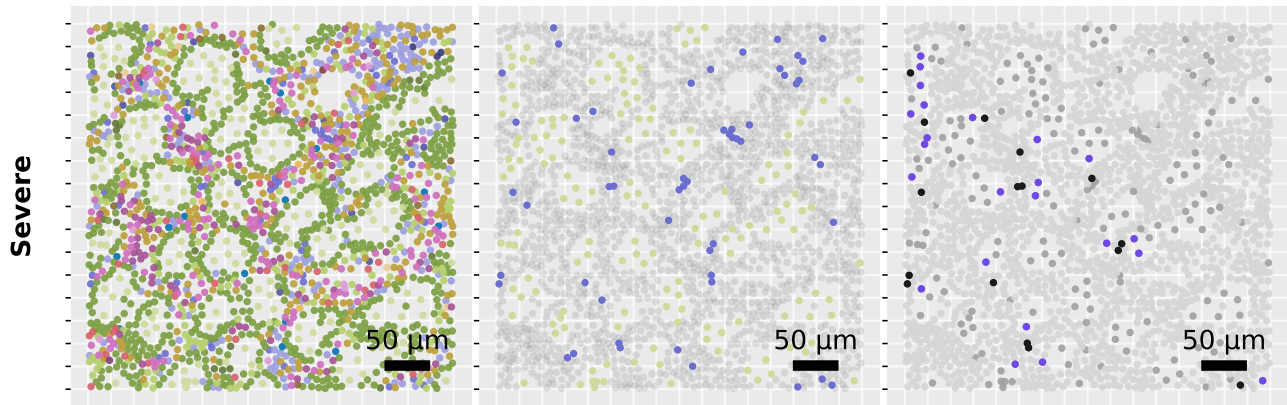
14 centers · 31/2443 cells



GVHD_14_FOV_3

lift 0.473 · conf 0.296 · conv 0.531
sup 0.00803 · lev -0.00894 · fdr 0.00569

16 centers · 39/2000 cells



Cell type

● CD4T ● Goblet

● center (CD4T) ● consequent (Goblet)

Figure 29: Maps and example selection follow Figure 17.

Muscle → CD4T avoidance rises from 4/18 to 17/25 eligible fields, showing increasing immune–wall separation.

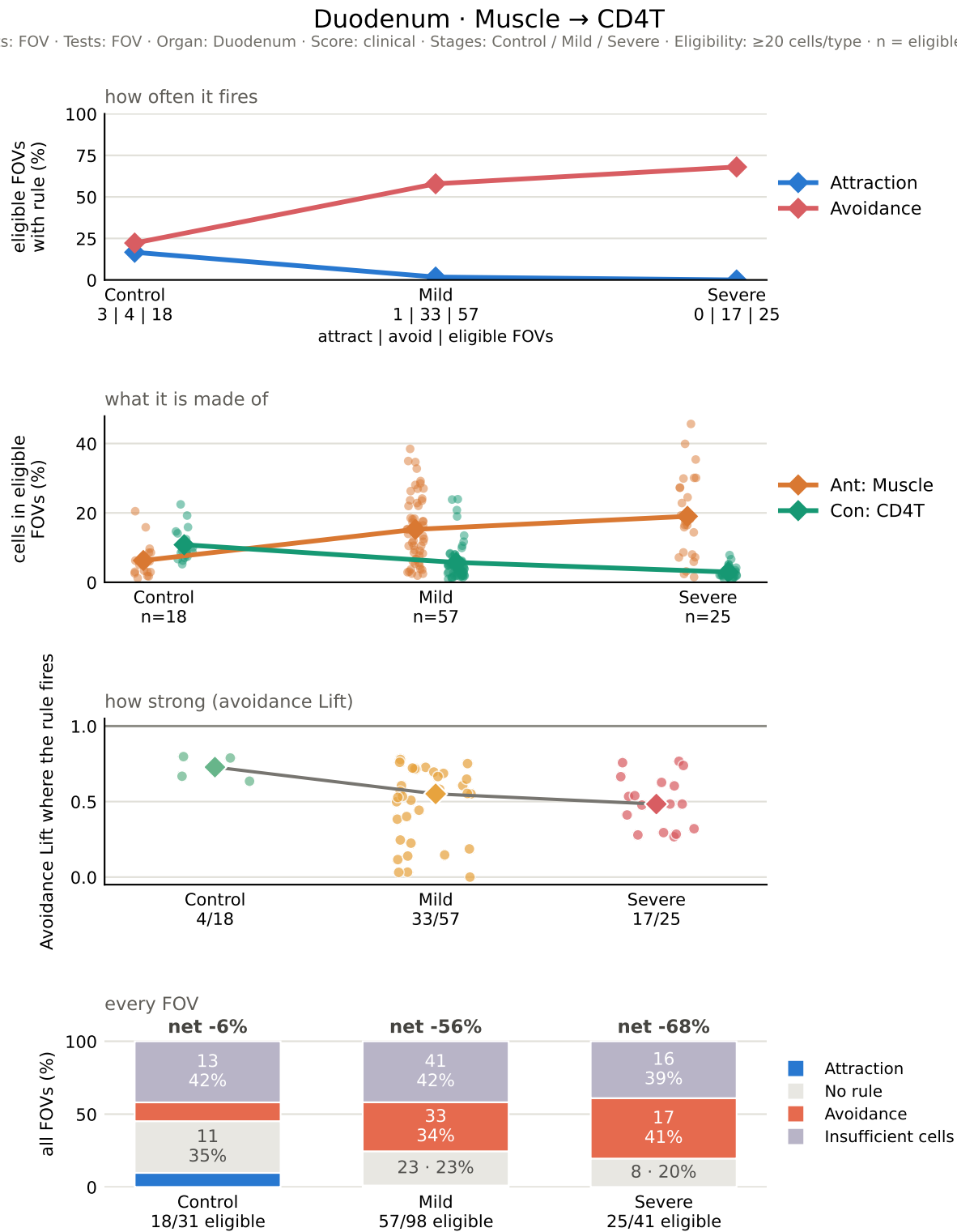


Figure 30: Read as in Figure 3. The Control–Severe avoidance gap is 46 percentage points; the rule occurs in 3/8 versus 7/9 eligible patients. This and the next CD8T result remain exploratory.

The highlighted fields place CD4 T cells away from muscle-rich regions where avoidance is mined.

Duodenum · Muscle → CD4T

Avoidance examples

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

rule cells

full FOV
Control_09_FOV_3

lift 0.789 · conf 0.545 · conv 0.679
sup 0.0509 · lev -0.0136 · fdr 0.00724

cells counted for the rule
103 centers · 133/2065 cells

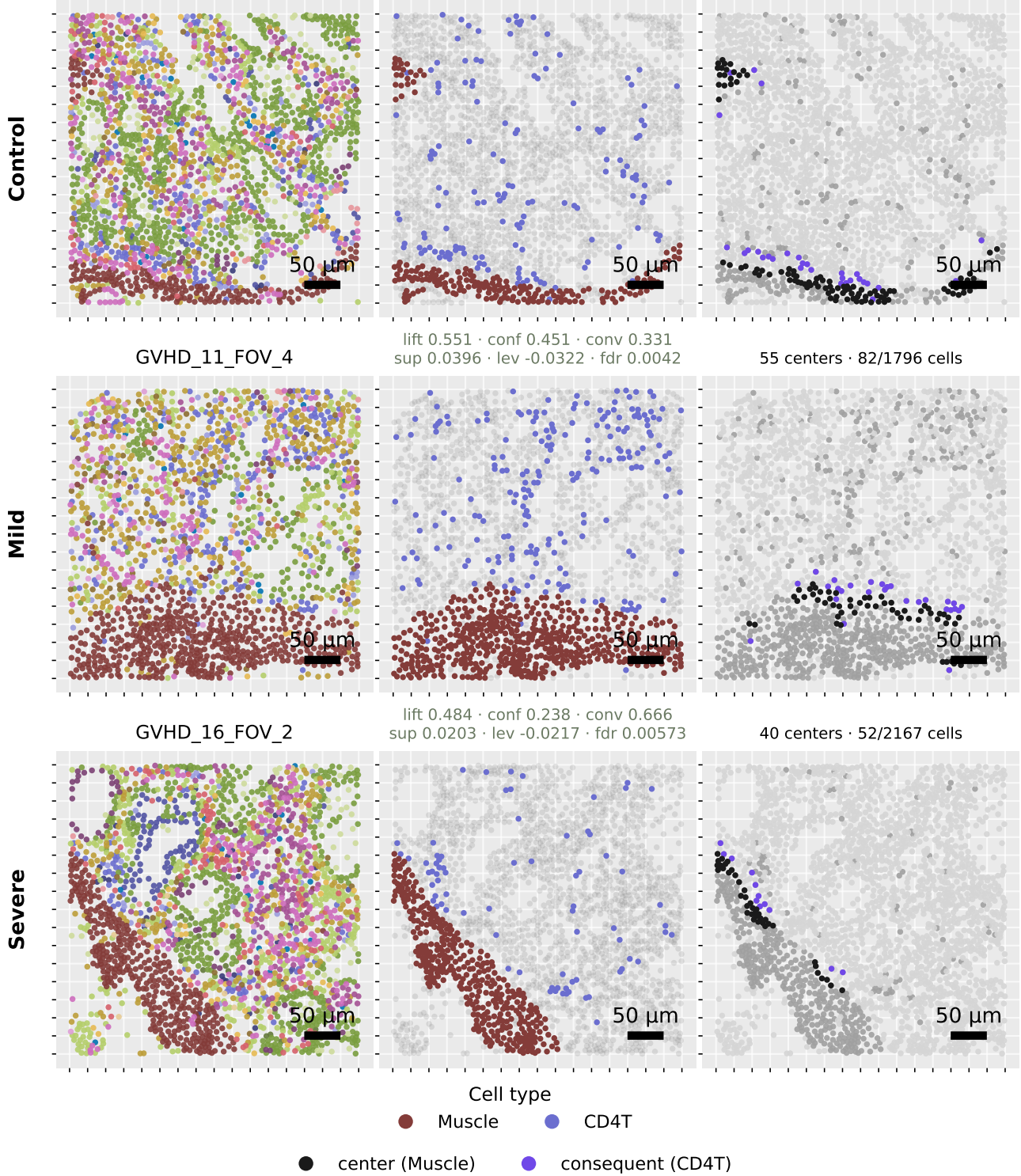


Figure 31: Maps and example selection follow Figure 17.

Muscle → CD8T avoidance also rises, from 6/17 Control to 16/22 Severe eligible fields, extending the immune-wall pattern to a second T-cell class.

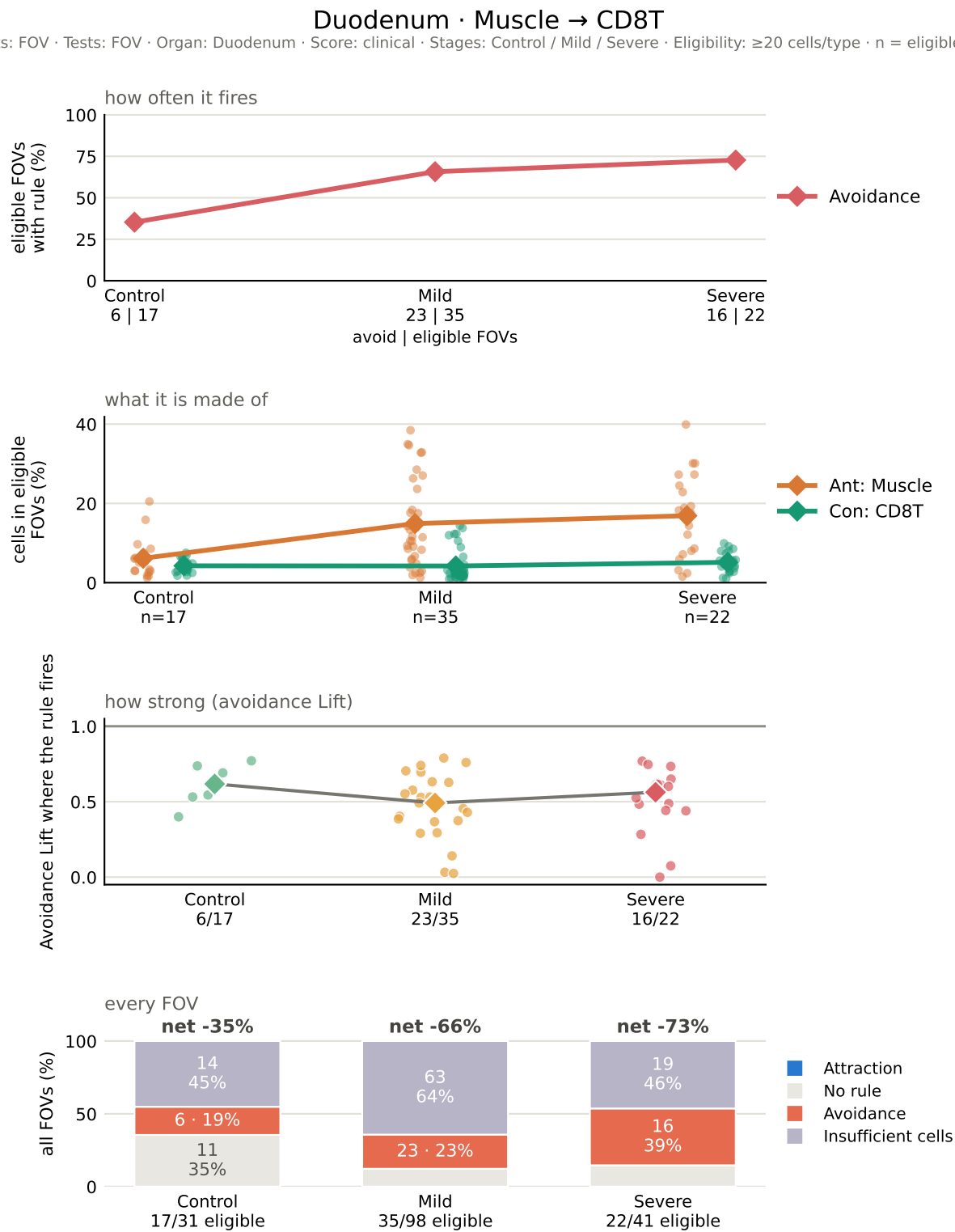


Figure 32: Read as in Figure 3. CD8T share among eligible fields is similar at both endpoints. The rule occurs in 6/7 versus 6/8 eligible patients, so the FOV increase does not correspond to more patients with any occurrence.

The highlighted fields show the muscle and CD8 T-cell positions behind this avoidance.

Duodenum · Muscle → CD8T

Avoidance examples

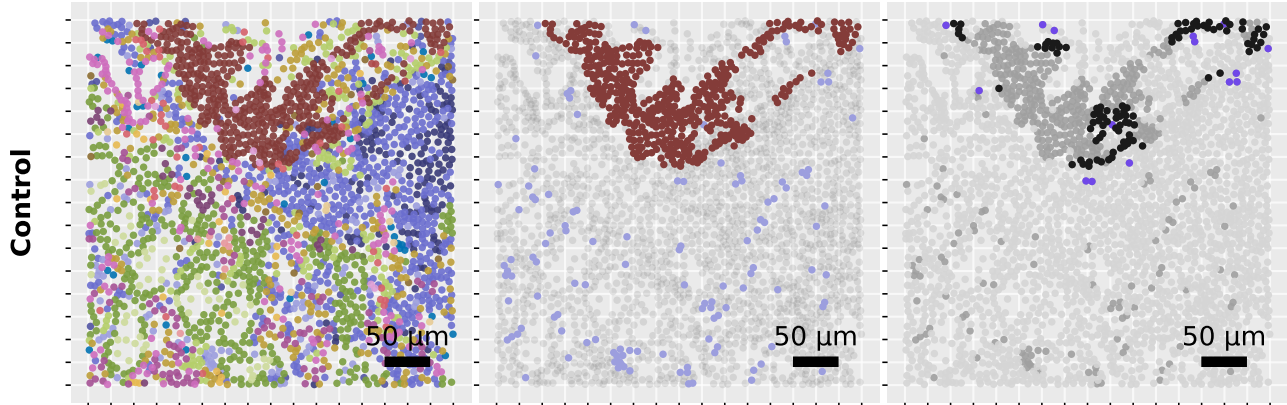
Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

rule cells

full FOV
Control_01_FOV_1

lift 0.544 · conf 0.326 · conv 0.596
sup 0.0388 · lev -0.0325 · fdr 0.0026

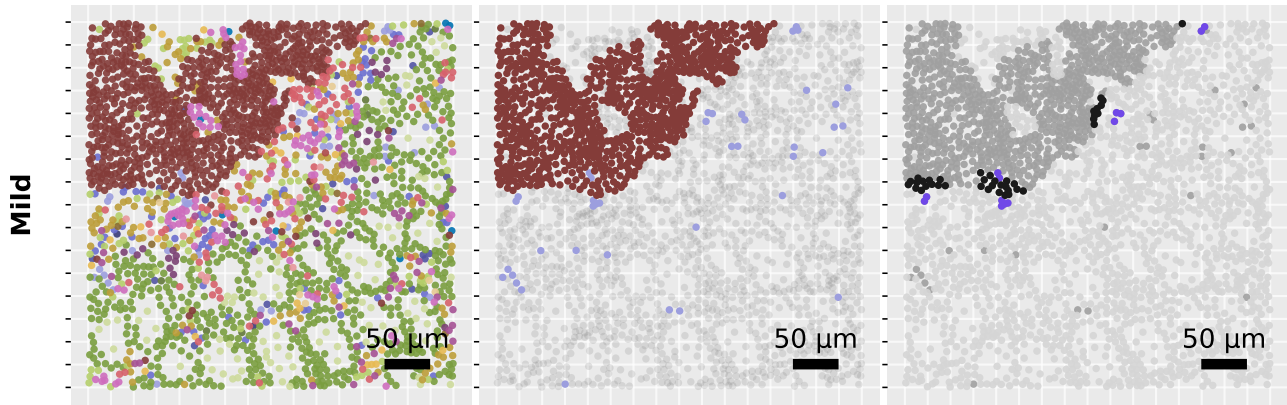
cells counted for the rule
97 centers · 112/2618 cells



GVHD_36_FOV_2

lift 0.491 · conf 0.145 · conv 0.825
sup 0.0218 · lev -0.0226 · fdr 0.0049

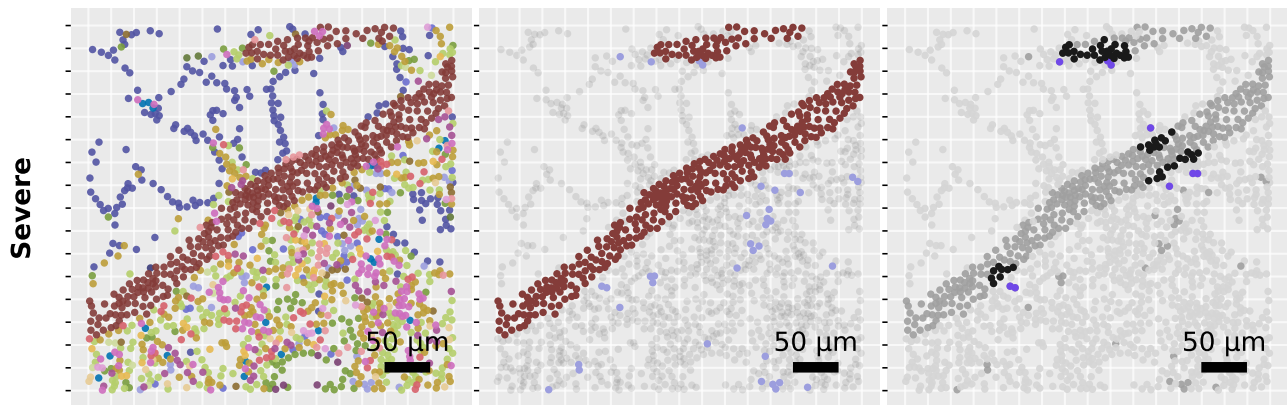
40 centers · 53/2409 cells



GVHD_24_FOV_3

lift 0.6 · conf 0.192 · conv 0.841
sup 0.0388 · lev -0.0258 · fdr 0.00296

56 centers · 65/1589 cells



Cell type
● Muscle ● CD8T
● center (Muscle) ● consequent (CD8T)

Figure 33: Maps and example selection follow Figure 17.

3.3 Fibroblast–muscle separation changes in opposite directions by organ

In colon, Fibroblast → Muscle avoidance falls from 13/20 Control to 9/28 Severe eligible fields.

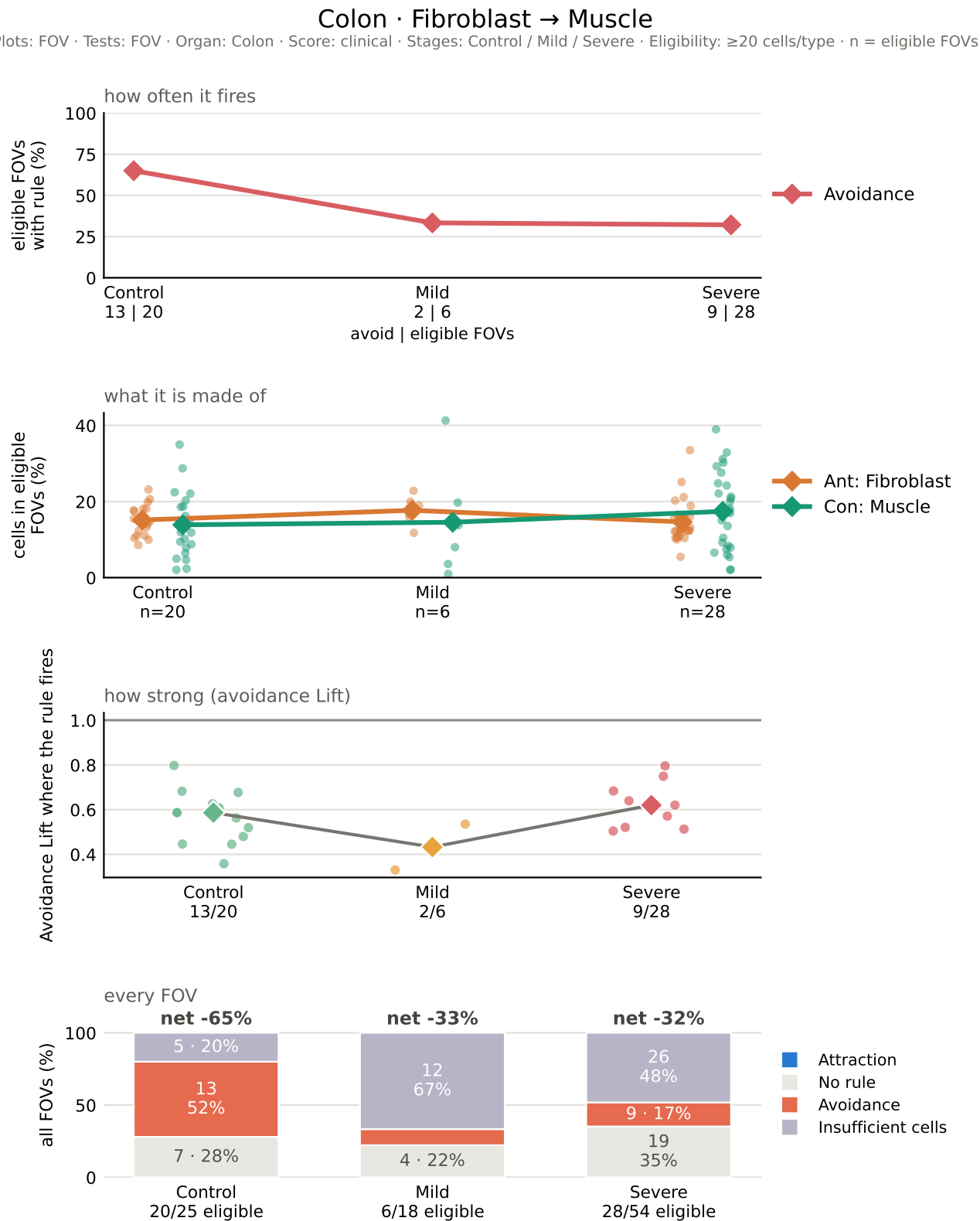


Figure 34: Read as in Figure 3. The 33-point Control–Severe decrease contrasts with the increase in duodenum below. The rule occurs in 4/4 Control versus 5/11 Severe eligible patients. Four Control patients and only 6 eligible Mild FOVs limit inference.

The colon examples show where fibroblasts and muscle produce the avoidance rule.

Colon · Fibroblast → Muscle

Avoidance examples

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

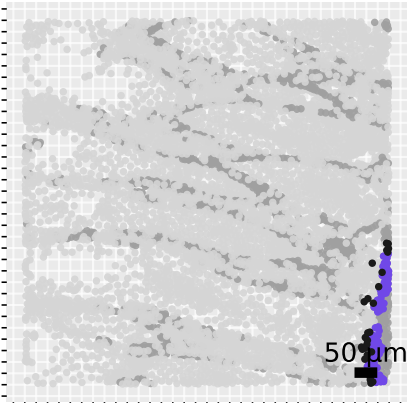
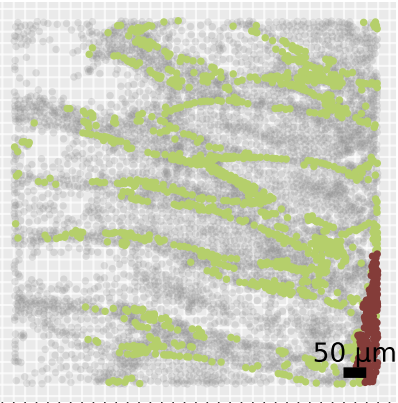
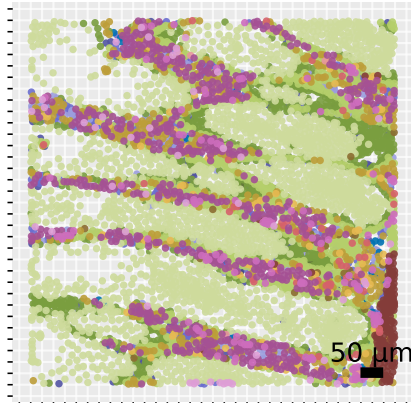
rule cells

full FOV
Control_Colon_03_FOV_6

lift 0.587 · conf 0.0267 · conv 0.981
sup 0.00559 · lev -0.00394 · fdr 0.00213

cells counted for the rule
28 centers · 113/6036 cells

Control

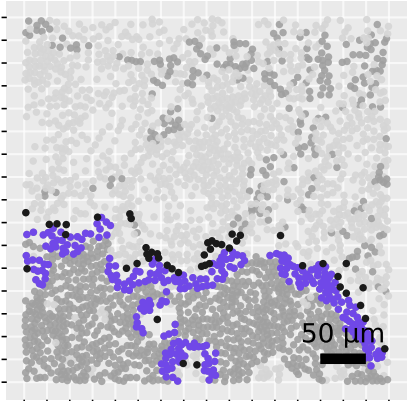
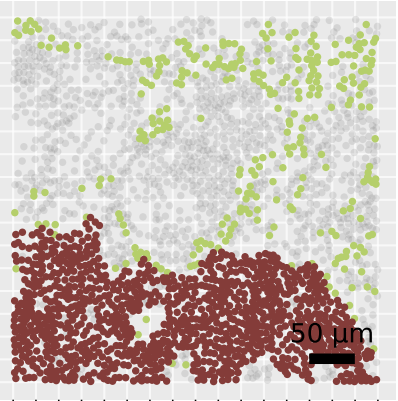
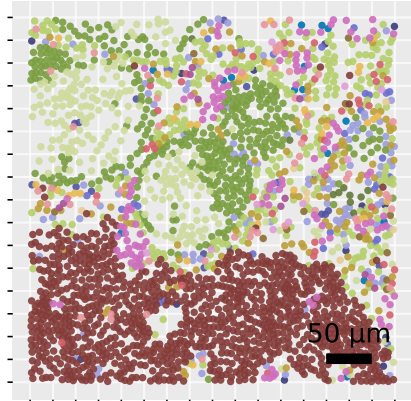


GVHD_21_FOV_3

lift 0.535 · conf 0.156 · conv 0.839
sup 0.0274 · lev -0.0238 · fdr 0.0027

47 centers · 315/2570 cells

Mild

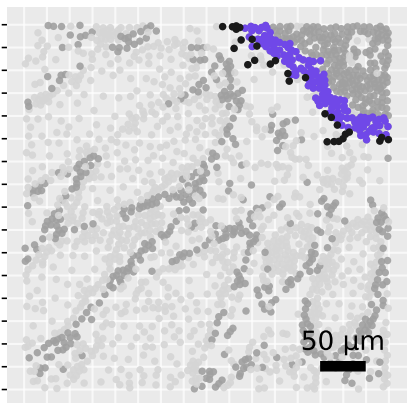
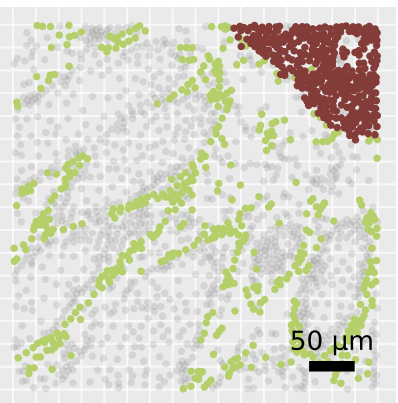
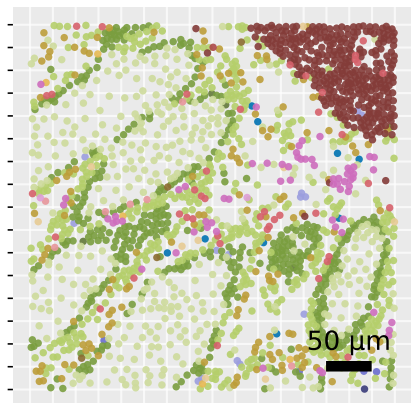


GVHD_59_FOV_3

lift 0.62 · conf 0.0561 · conv 0.964
sup 0.017 · lev -0.0104 · fdr 0.00252

28 centers · 152/1993 cells

Severe



Cell type

● Fibroblast ● Muscle

● center (Fibroblast) ● consequent (Muscle)

Figure 35: Maps and example selection follow Figure 17.

In duodenum, Fibroblast → Muscle avoidance instead rises from 1/18 Control to 17/28 Severe eligible fields, a 55-point increase.

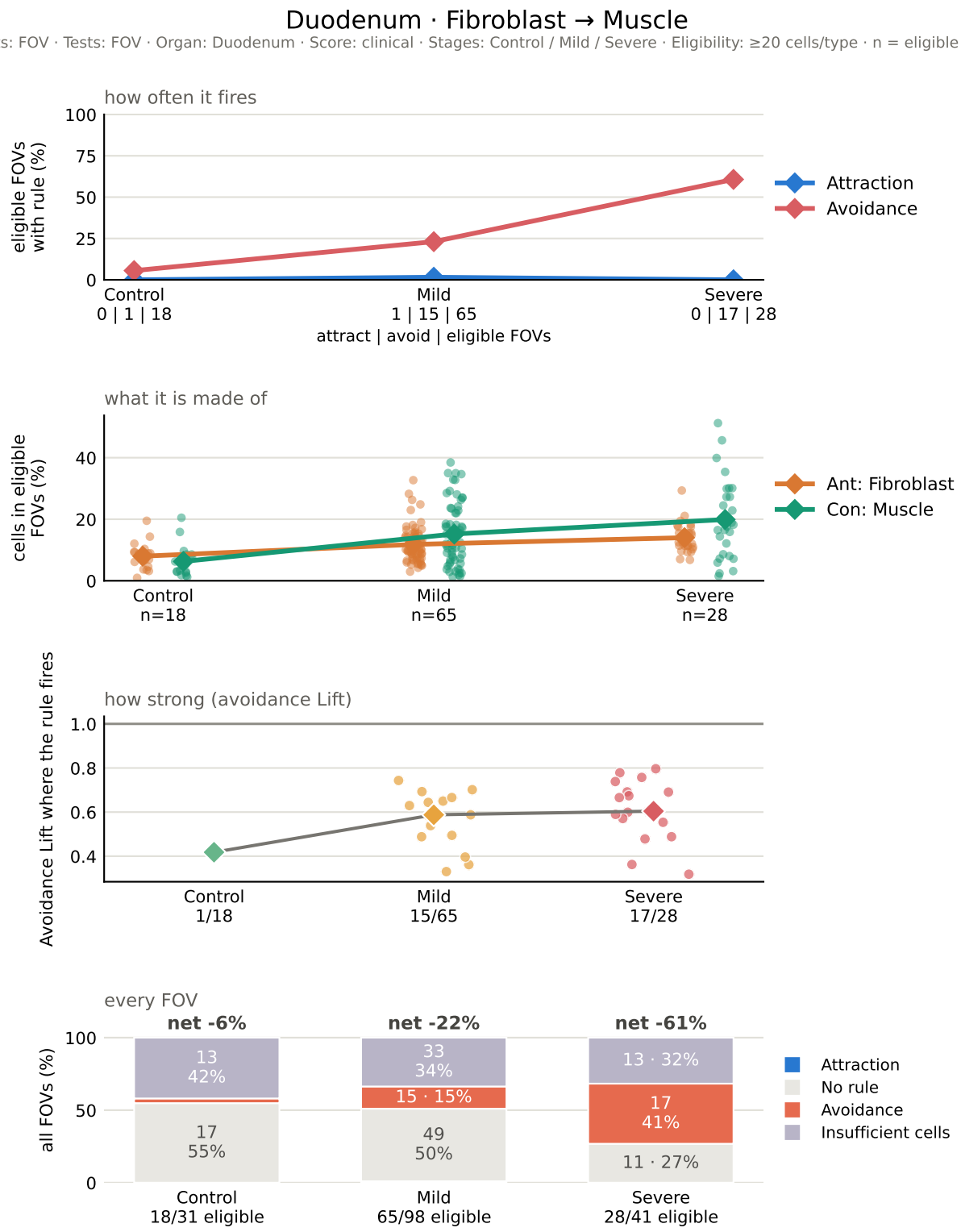


Figure 36: Read as in Figure 3. The rule occurs in 1/8 Control versus 7/9 Severe eligible patients. The same automatic FOV screen found the opposite colon pattern; both comparisons remain exploratory. Human intestinal GVHD has documented myofibroblast engraftment [7]; this provides context, not confirmation of this spatial trend.

The highlighted fibroblasts and muscle show the compartments behind duodenal avoidance.

Duodenum · Fibroblast → Muscle

Avoidance examples

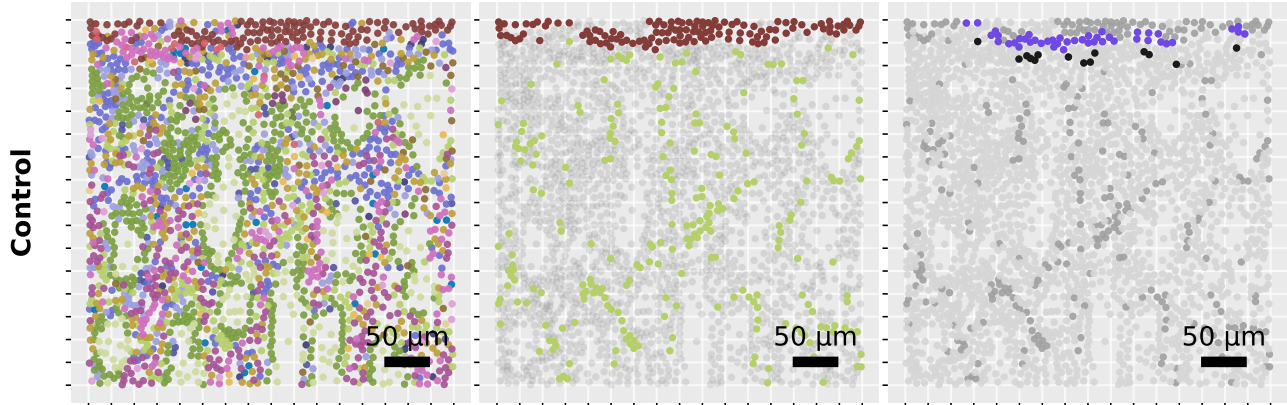
Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

rule cells

full FOV
Control_04_FOV_1

lift 0.418 · conf 0.0591 · conv 0.913
sup 0.00615 · lev -0.00856 · fdr 0.00292

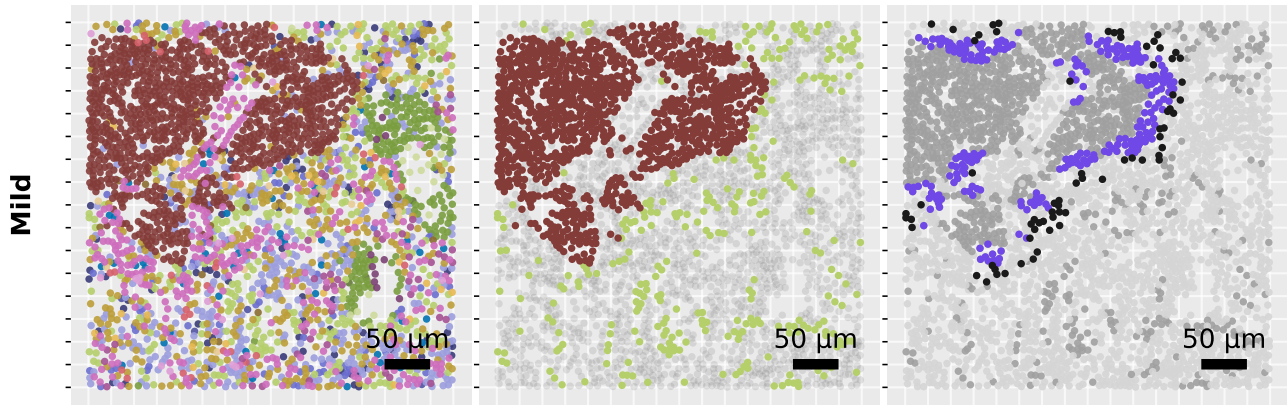
cells counted for the rule
14 centers · 58/2290 cells



GVHD_27_FOV_3

lift 0.588 · conf 0.227 · conv 0.794
sup 0.0271 · lev -0.019 · fdr 0.00208

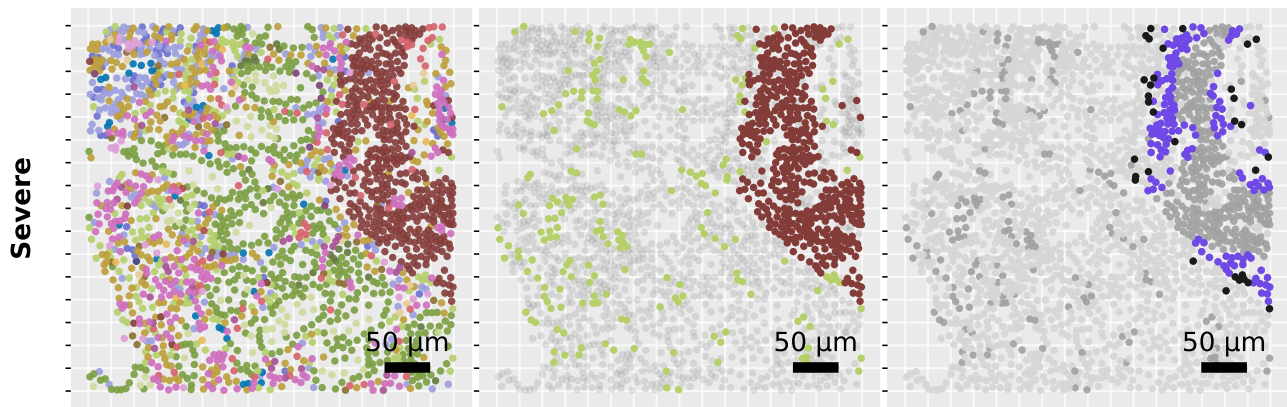
69 centers · 308/3061 cells



GVHD_14_FOV_1

lift 0.604 · conf 0.155 · conv 0.88
sup 0.0156 · lev -0.0102 · fdr 0.00369

32 centers · 173/2206 cells



Cell type

● Fibroblast ● Muscle

● center (Fibroblast) ● consequent (Muscle)

Figure 37: Maps and example selection follow Figure 17.

3.4 Pair rules also change over time after transplantation

The atlas motivates the time windows through immune-cell kinetics [1]. The post-transplant screen compares rule prevalence in three windows, with Control shown only as context. Time and disease severity can vary together, so these patterns are exploratory.

Colon Macrophage → CD4T attraction falls from 7/16 eligible fields before day 30 to 0/11 after day 100.

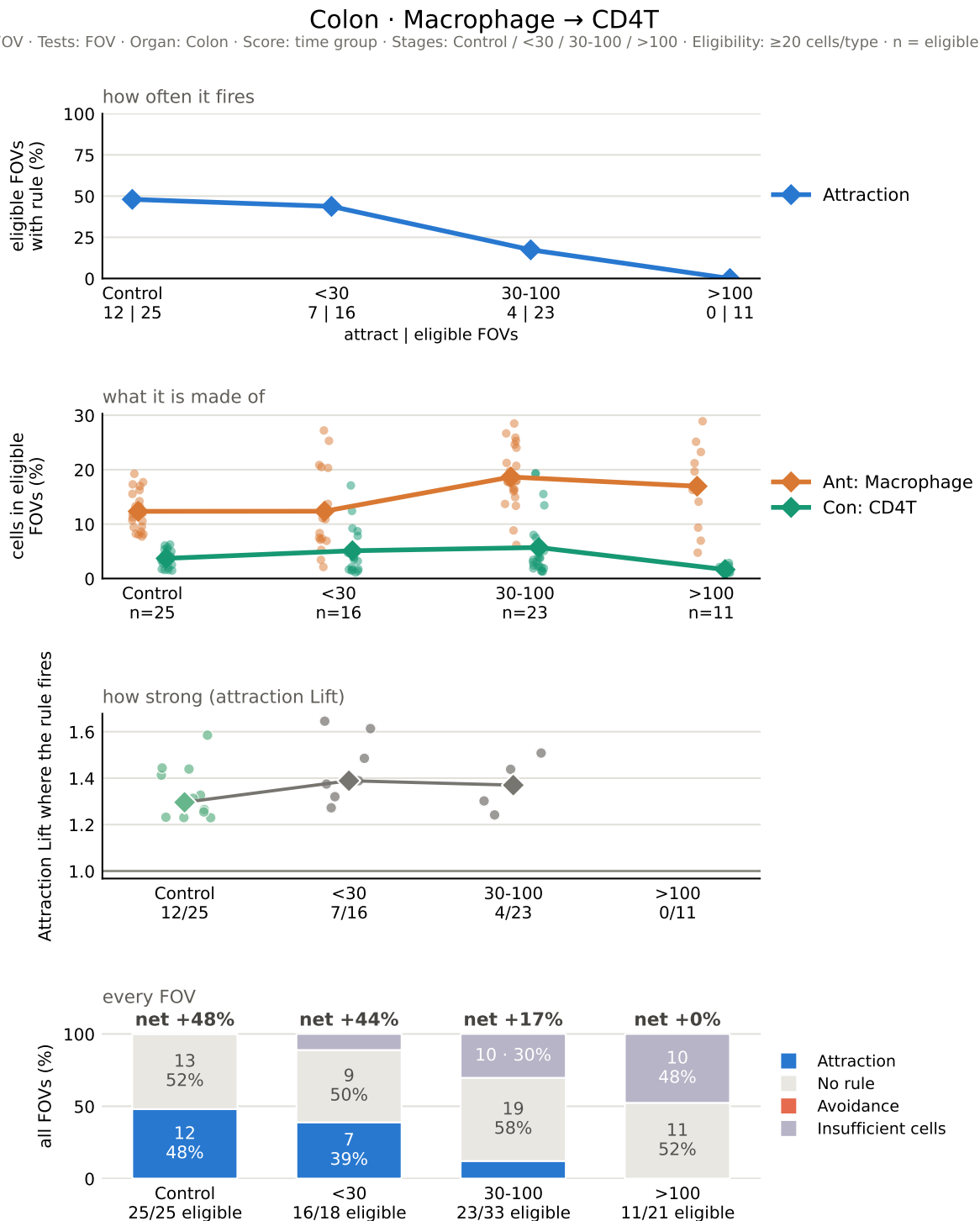


Figure 38: The four rows follow Figure 3, with time windows replacing clinical stages. The rule appears in 4/5 eligible early patients and 0/4 eligible late patients; the 44-point FOV gap has no adjusted patient-level test.

The early highlighted fields show where Macrophage → CD4T attraction fires.

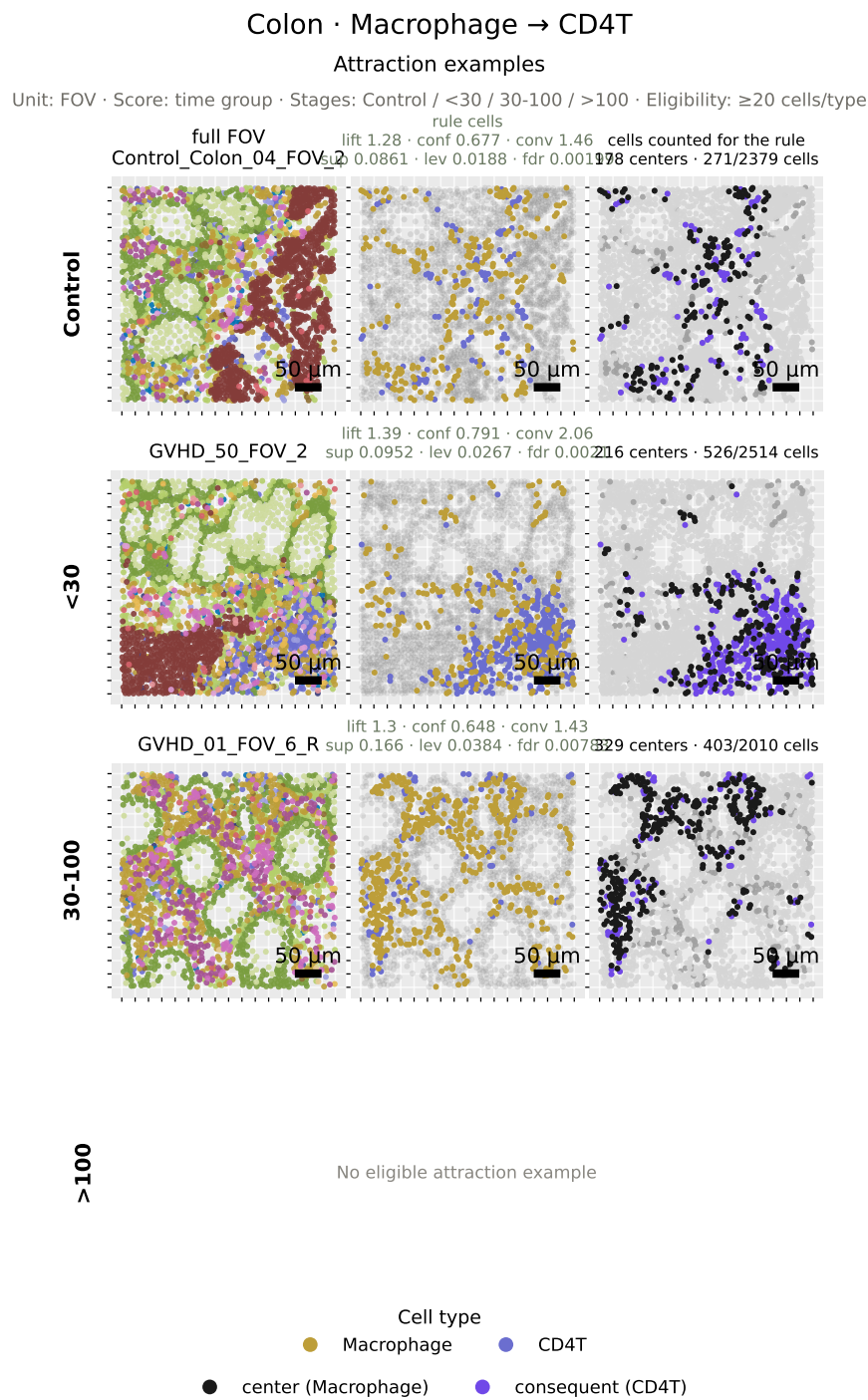


Figure 39: Maps and example selection follow Figure 17.

Later eligible fields without the rule show the contrast to those early examples.

Colon · Macrophage → CD4T

Fields without the rule

Unit: FOV · Score: time group · Stages: Control / <30 / 30-100 / >100 · Eligibility: ≥ 20 cells/type

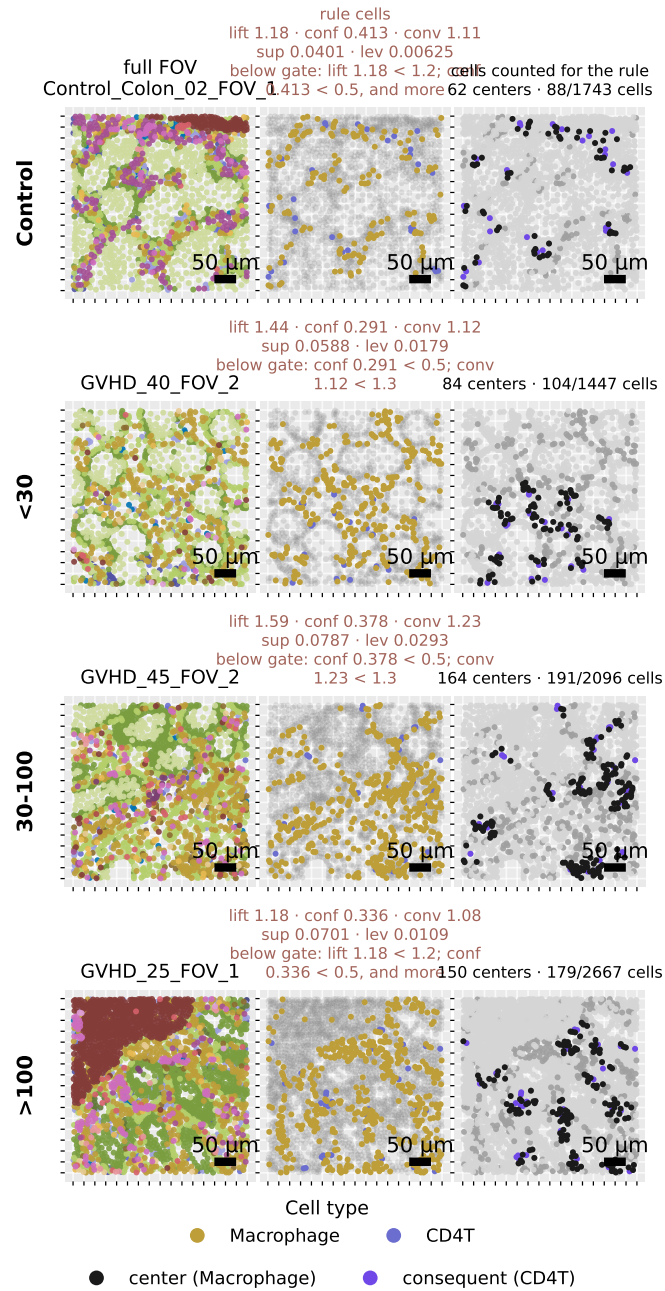


Figure 40: Maps and example selection follow Figure 17.

Duodenal Plasma_CD38 → Fibroblast attraction rises from 3/60 eligible fields before day 30 to 11/25 after day 100.

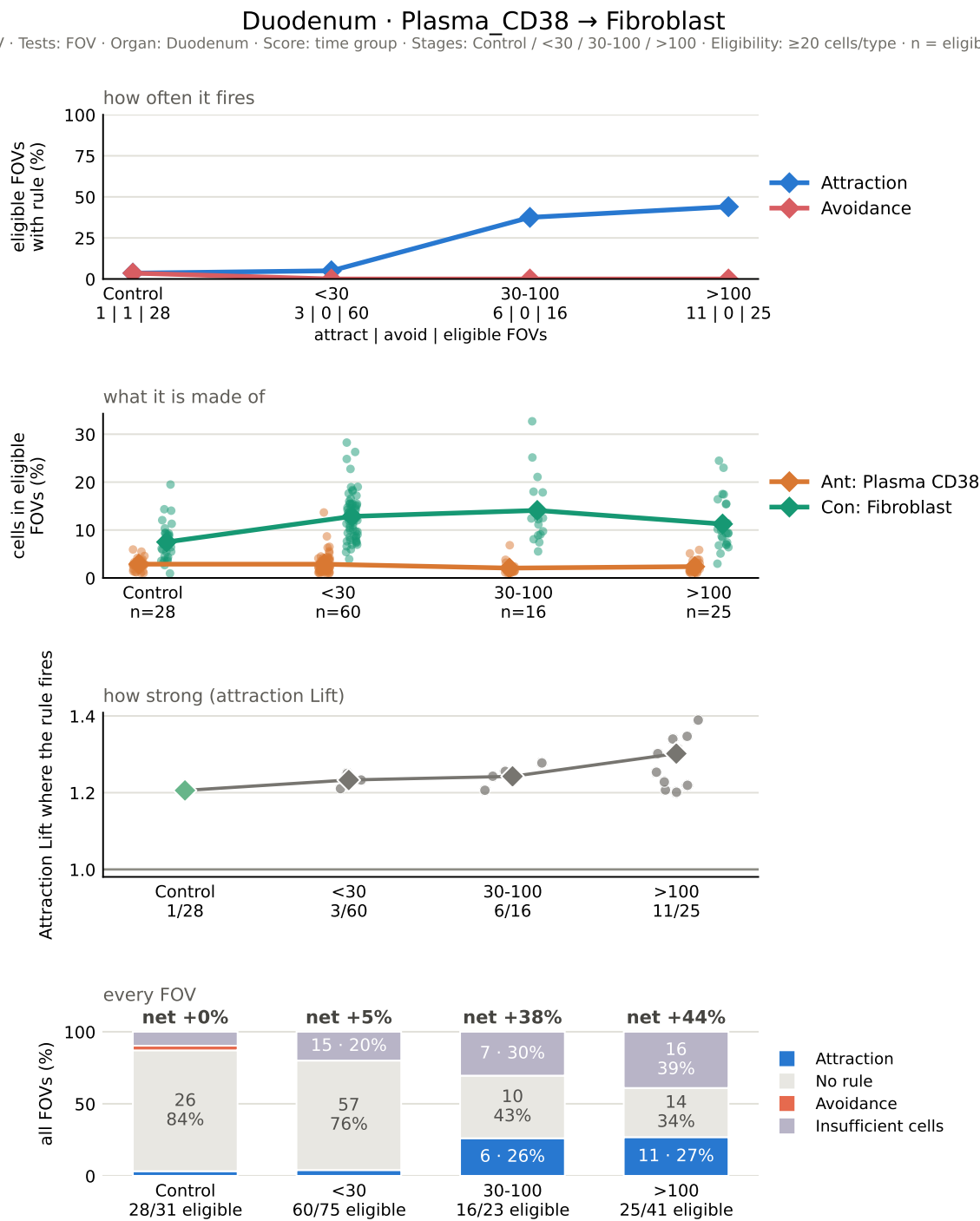


Figure 41: Read as in the preceding time figure. The rule appears in 3/21 eligible early patients and 5/8 eligible late patients; the 39-point FOV gap remains exploratory.

The highlighted tissue examples show where the later Plasma_CD38–fibroblast rule fires.

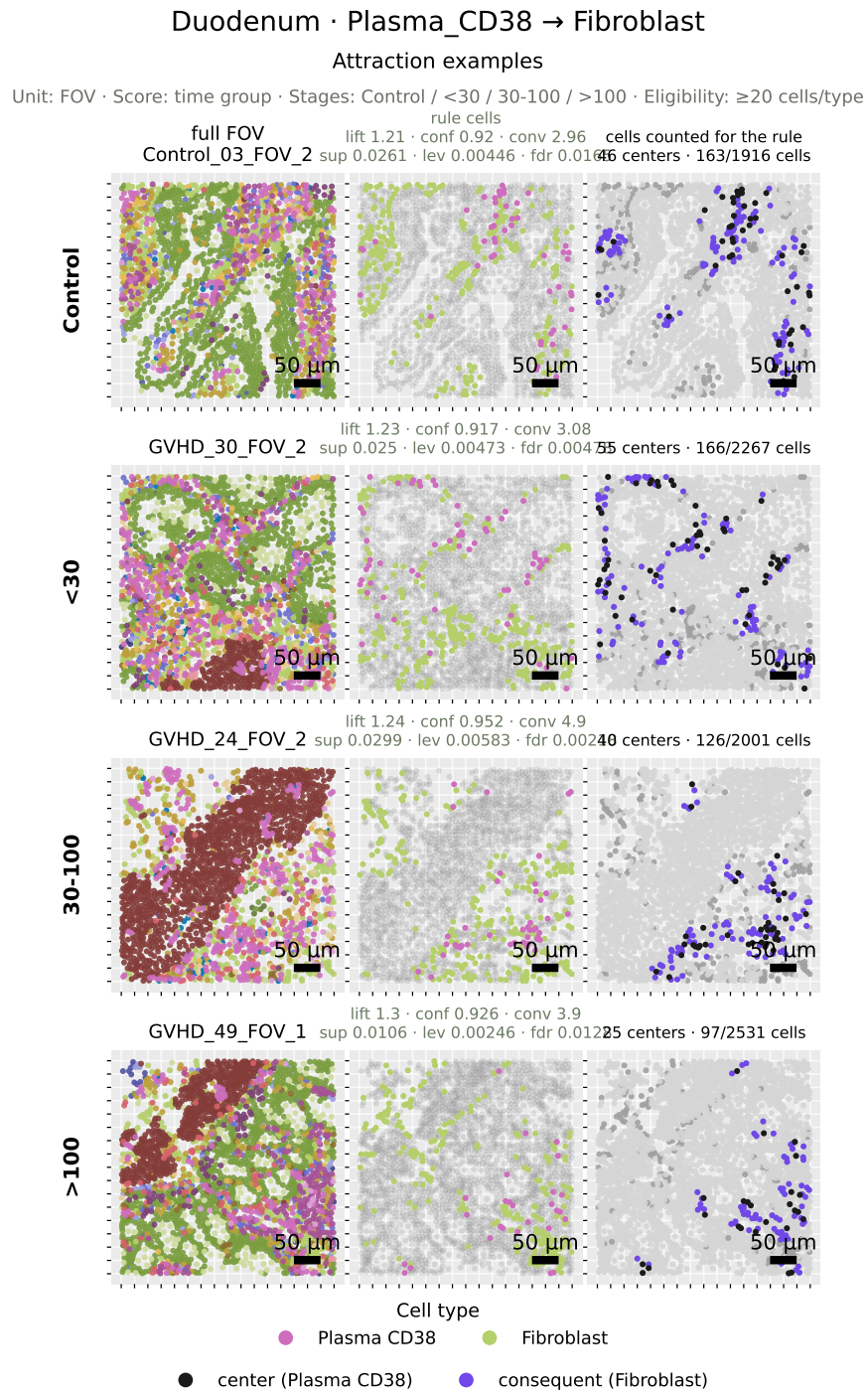


Figure 42: Maps and example selection follow Figure 17.

Eligible fields without the rule show how these cell types appear outside that arrangement.

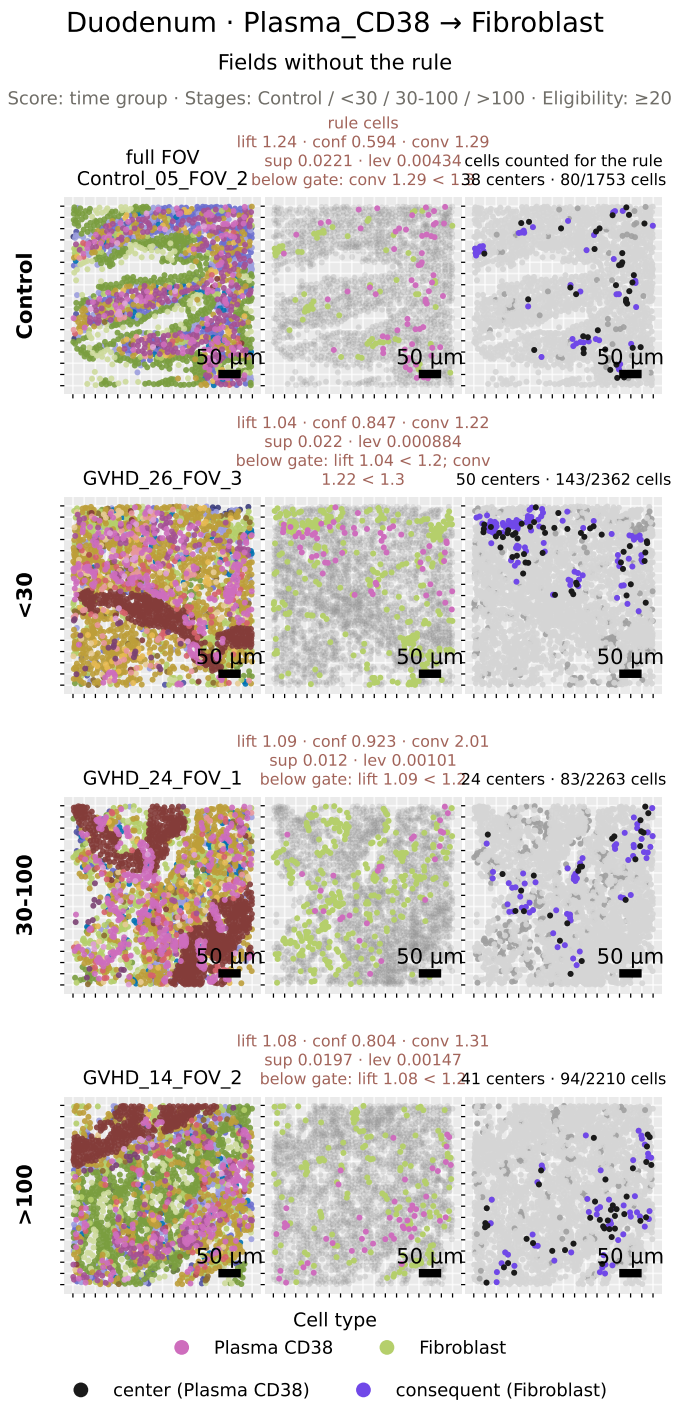


Figure 43: Maps and example selection follow Figure 17.

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